

OSPREY: HIGH-SPEED ELECTRIC UNMANNED SURFACE VESSEL FOR PAYLOAD TRANSPORT

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FINAL REPORT

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Abstract

The US Navy is always exploring various manned and unmanned vessel options for its fleets and aims to develop its workforce to continue this effort. This thesis presents the design, assembly, and validation of Osprey: a high-speed, unmanned surface vessel for payload transportation to compete in the Promoting Electric Propulsion competition run by the American Society of Naval Engineers (ASNE) and the Office of Naval Research (ONR). The vessel runs twin inboard motors capable of outputting 12 kW of power at 44.4 V nominal, with a total weight of 120 lbs and a theoretical top speed of 80 mph. The hull is fully custom-made in-house using a carbon fiber resin infusion process. The powertrain features a water-cooled heat-exchanging system and a conventional RC-scaled boat driveline setup. The electrical system includes a low voltage battery and a remote kill switch module that allows the vessel to be shut off at any time. Dry and wet testing validated that the vessel meets all design requirements and began to provide an understanding of the vessel's on-water performance, demonstrating that Osprey is an excellent, reliable platform to continue developing and testing.

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To PES.

Contents

Abstract	iii
Acknowledgements	iv
List of Tables	viii
List of Figures	ix
List of Symbols	xiii
1 Problem Definition	1
1.1 Motivation	1
1.2 Scope of Work	2
2 Engineering Design	3
2.1 Iterative Process	3
2.2 Engineering Standards	4
2.3 Design Requirements and Constraints	4
3 Design	5
3.1 Hull	5
3.1.1 Base Model	5
3.1.2 Optimized Scaling	7
3.2 Powertrain	20
3.3 Driveline	21
3.4 Steering	24
3.4.1 Component Selection	24
3.4.2 Rigging Design	25
3.5 Coolant System	28
3.6 Component Mounting	30
3.6.1 Battery Rails	30
3.6.2 General Component Mounting	31
3.7 Electrical System	32

3.7.1	High-Voltage Propulsion System	33
3.7.2	Low-Voltage Auxiliary and Control System	34
4	Manufacturing	35
4.1	Hull	35
4.1.1	Molds	35
4.1.2	Resin Infusion	43
4.1.3	Flat Stock	48
4.1.4	Post Processing	49
4.1.5	Assembly	52
4.2	Driveline Assembly	54
4.3	Component Mounts	55
4.4	Steering Assembly	55
4.5	Low-Voltage Box	56
5	Integration	57
5.1	Mounting	58
5.2	Coolant	60
5.3	Steering	60
6	Design Evaluation	62
6.1	Testing	62
6.2	Post-Assembly Challenges	64
7	Conclusion & Future Work	68
7.1	Progress Summary and Assessment	68
7.2	Recommendations for Future Work	69
A		74
A.1	MATLAB Scripts	74
A.1.1	Hull Sizing Script	74
A.1.2	aeroForce Function	78
A.1.3	calcWettedAreaIterative Function	79

List of Tables

2.1	System-Level Design Requirements, Rationale, and Validation Methods	4
3.1	Initial Assumptions, Fixed Geometry, and Designer-Selected Variables for the Final Hull Design	16
3.2	Estimated Weight Breakdown of the Boat by Subsystem	17
3.3	Derived Geometry and Predicted Performance for the Final Hull Design	18

List of Figures

3.1	Mystic C5000	6
3.2	Initial Mystic C5000 CAD Model	6
3.3	Mystic C5000 CAD Side View	7
3.4	Mystic C5000 CAD Model Views	7
3.5	<i>Secrets of Tunnel Boat Design</i> Graph 1: Lift Coefficient Air, C_{LA} . .	8
3.6	<i>Secrets of Tunnel Boat Design</i> Graph 2: Drag Coefficient Air, C_{DA} .	10
3.7	<i>Secrets of Tunnel Boat Design</i> Graph 3: Lift Coefficient Water, C_{LW}	11
3.8	<i>Secrets of Tunnel Boat Design</i> Graph 4: Drag Coefficient Water, C_{DW}	13
3.9	<i>Secrets of Tunnel Boat Design</i> Graph 5: Motor Drag, D_M	14
3.10	Final Scaling CAD	19
3.11	Final Scaling Front View	19
3.12	Final Scaling Rear View	19
3.13	Motor, ESC, and Battery Selection	20
3.14	Driveline Components	21
3.15	Pipe Bending Jig	22
3.16	Driveline Plates Installed in Sponson	23
3.17	Driveline Plates	23
3.18	Steering Components	24
3.19	Servo Mount	25
3.20	Steering Cable Router	26
3.21	Rudder Mount	26
3.22	Rudder Block	27
3.23	Steering Assembly CAD	27
3.24	Cooling System Components	29
3.25	Battery rails with battery and payload mounts	31
3.26	Battery Rail Slider Nut	31
3.27	Mounting styles for general component mounting	32
3.28	KiCad schematic of the LV electrical system	33

4.1	Plane where mold was cut in half	35
4.2	Flat Sponson Edge (Highlighted in blue)	36
4.3	Bottom Half Mold CAD	37
4.4	Bottom Half Mold CAD Bottom View	37
4.5	Top Half Mold CAD	38
4.6	Top Mold CAD Bottom View	38
4.7	Example Mold Piece	39
4.8	Assembled Bottom Half Mold	40
4.9	Top and bottom molds assembled with 2x4 reinforcement	41
4.10	Top and bottom molds primed and with flange extension (photo taken after infusion part removal, so flange appears damaged)	41
4.11	Canopy Mold CAD	42
4.12	Completed Canopy Mold	42
4.13	Carbon Laid Bottom Half Mold	44
4.14	Carbon Laid Canopy Mold	44
4.15	Fully layered molds prior to vacuum bagging	45
4.16	Bottom Mold Under Vacuum	46
4.17	Infused molds after resin infusion	47
4.18	Bottom half being debugged	47
4.19	Finished composite parts in molds	47
4.20	Flat Stock Infusion	48
4.21	Flat Stock	48
4.22	Water being siphoned under the carbon to release part	49
4.23	Freshly removed composite parts	50
4.24	Bare Assembly	50
4.25	Deck Door Integration and Removal	51
4.26	Assembled hull with deck doors cut	51
4.27	Bulkhead Installation and Preparation	52
4.28	Float test of bottom half	53
4.29	Top half gluing down to bulkheads	53
4.30	Fully assembled hull with seam tape	54
4.31	LV Box	56
5.1	Fully Integrated System (No Payload)	57
5.2	Final Weigh In	58
5.3	Battery Rails Installed	59

5.4	Both styles of mounting brackets installed	59
5.5	Water pickup (left) and coolant output nipple (right)	60
5.6	Rudder Mount Integration	61
5.7	Integrated Steering Assembly	61
6.1	Castle Link Data Log Page	62
6.2	Load Testing in Water	63
6.3	Full System Test Run	63
6.4	Planing Run Data Log	64
6.5	ESC after fire	66

List of Symbols

L_A	Aerodynamic Lift	8
ρ	Air Density	8
V	Top Speed	8
S	Tunnel Reference Area	8
C_{LA}	Coefficient of Aerodynamic Lift	8
$CLA1$	Base coefficient of aerodynamic lift before modification	8
α_{wing}	Angle of attack of the tunnel	8
hc	Height Ratio	8
H	Tunnel height at the back of the tunnel	8
c	Chord Length	8
α_{trim}	Angle of attack of the bottom of the hull	8
α_{aero}	Angle of attack of the tunnel with respect to the bottom of the hull	8
AR	Tunnel Aspect Ratio	9
TC	Tunnel Thickness Ratio	9
FAR	CLA1 modifier for aspect ratio	9
FTC	CLA1 modifier for thickness ratio	9
D_A	Aerodynamic Drag	9
S_{avg}	Average Reference Area	9
C_{DA}	Coefficient of Aerodynamic Drag	9
CDI	Coefficient of Induced Aerodynamic Drag	10
CDO	Coefficient of Friction Drag	10
L_W	Hydrodynamic Lift	10
W_{tot}	Total Weight	10
S_W	Wetted Sponson Area	11
C_{LW}	Coefficient of Hydrodynamic Lift	11
$\alpha_{sponson}$	Sponson Angle of Attack	11
A_W	Wetted Aspect Ratio	11
α_{hydro}	Local Step Angle	11

$CLWb$	Base coefficient of hydrodynamic lift before modification	11
FDR	$CLWb$ modifier for deadrise angle	11
l_W	Wetted Length	12
b	Sponson Width	12
D_W	Hydrodynamic Drag	12
ρ_W	Water Density	12
C_{DW}	Coefficient of Hydrodynamic Drag	12
CDF	Coefficient of friction drag due to water	13
$CDF1$	Base coefficient of friction drag without modification	13
SAF	Spray Area Factor	13
vl	Velocity Length	13
CDI_W	Coefficient of induced drag due to water	13
CDP	Profile Drag	13
S_A	Cockpit cross-sectional area	14
$C_{D,cockpit}$	Coefficient of drag for cockpit	14
$D_{cockpit}$	Appendage Drag	14
D_{tot}	Total Drag	15
P_{req}	Power Required	15
P_{avail}	Power Available	15
A_{rudder}	Wetted area of rudder	24
$C_{D,rudder}$	Coefficient of drag on rudder	24
Q	Heat Loss	29
P	Input Power	29
η	Total Efficiency	29
$\dot{m}_2$	Mass flow rate of raw water pickup	29
c_p	Specific heat of brackish water	29
v	Velocity	30
A_{tube}	Area of water pickup	30
A	Surface area of heat exchanger	30
α	Heat transfer coefficient for heat exchanger	30

Chapter 1

Problem Definition

1.1 Motivation

The Promoting Electric Propulsion competition (PEP), run by the American Society of Naval Engineers (ASNE) and the Office of Naval Research (ONR), hosts a 2-mile race won by a vessel carrying a 30-lb payload with the fastest finishing time. Running in brackish water with unpredictable conditions, a vessel must be developed with proper energy capacity, power output, and hull performance to succeed. This project directly tackles this challenge.

The Promoting Electric Propulsion website explains the relevance of the competition to the development of Navy vessels: “Today’s Navy is exploring a variety of crewed and uncrewed platforms. It has been exciting to see a US-built autonomous fleet journey from San Diego to Australia (USNI News), and the Naval Research Enterprise (NRE) is rapidly expanding the number of new platforms and systems. For example, the Defense Innovation Unit sought proposals on scalable Small Unmanned Surface Vessels that can carry 1,000 pounds of fuel while operating in contested environments for days (DefenseScope). It’s an amazing time when small uncrewed craft are 11 meters, medium USVs are anything less than 200 feet, and large unmanned surface vessels (LUSVs) are growing in size and complexity.” [1].

A vessel fit for PEP can serve a variety of purposes. The payload could be anything from explosives to medical supplies, or even surveillance equipment. It could be used for both data collection and payload delivery.

1.2 Scope of Work

The scope of this thesis project will include hull design optimization and manufacturing, mechanical and electrical design and integration, and testing. Note RC control is not included in the scope. The development stage includes hull calculations, selection of major powertrain and driveline components, modeling of vessel layout, and design of component mounts. The manufacturing stage includes production of hull molds, carbon fiber resin infusion process, prototyping and final printing of 3D-designed parts, and wiring of electrical system. The integration stage includes mounting all components into the hull. Finally, the evaluation stage includes dry testing of powertrain, electrical, and cooling systems, as well as on-water testing of leakage and hull/powertrain performance.

Chapter 2

Engineering Design

The engineering design process is described throughout the body of this paper. Chapter 3 covers the design process, Chapter 4 the manufacturing, and Chapter 5 the integration. The results of the finalized design and improvements to be made are discussed in Chapter 6 and beyond. Here is a summary of the engineering design process followed for this project.

2.1 Iterative Process

Step	Iterative Process
1	Initial proposal and concept development.
2	Powertrain selection and preliminary hull layout.
3	Electrical, propulsion, and RC system design.
4	Structural design and internal arrangement refinement.
5	Integration of propulsion, electrical, and RC systems.
6	Vessel dry and wet testing.
7	Design revisions based on test results and integration/assembly challenges.
8	Final design after repeated iteration, troubleshooting, and validation.

2.2 Engineering Standards

Engineering Standard	Description
ABYC E11	Guide for the design, construction, and installation of direct current (DC) electrical systems on boats that operate at potentials of 50 volts or less.

2.3 Design Requirements and Constraints

Requirement	Rationale	Validation Method
A main power disconnect must control all current to the propulsion system. When switched off, all high-voltage systems must cease operation.	Reduces risk of electrical shock and prevents unintended propulsion.	Design testing/integration
The battery enclosure must securely retain the battery within the vessel.	Ensures batteries remain contained and recoverable if the vessel capsizes.	Water propulsion test
The vessel must operate at ≤ 55.5 V and ≤ 500 Ah capacity.	PEP competition requirement.	Design integration
All battery current must pass through a properly rated fuse.	Protects electrical components from overcurrent.	Design testing/integration
A towing structure must be reachable by a boat hook.	Allows recovery or restraint by a safety vessel.	Design integration
The vessel must have positive buoyancy.	Ensures recovery if the vessel floods.	Water float test
The vessel must carry a 30 lb removable payload.	Demonstrates applicability for real-world missions.	Design integration

Table 2.1: System-Level Design Requirements, Rationale, and Validation Methods

Chapter 3

Design

3.1 Hull

3.1.1 Base Model

The key design requirements considered for the hull were its ability to handle the 30 lb payload and rough water while optimizing speed. A tunnel hull is the best option when trying to go fast due to its ability to produce aerodynamic lift. The fastest type of tunnel hull is a hydroplane, something we both have experience with through Princeton Electric Speedboating. However, hydroplanes have a very flat bottom and require extremely smooth water to run fast. As an RC boat scaled down, this weakness would only be exaggerated, eliminating the hydroplane as an option. Instead, we chose a traditional racing catamaran, which has a taller tunnel and can therefore handle rougher waters. Specifically, we based our design on the Mystic C5000, traditionally a 51-ft racing hull with just under 4000 HP (Figure 3.1) [2]. The Mystic C5000 model is also commonly used by hobbyists for larger RC boats, typically in the 6 ft range. The particular model we used has several features suitable for this project. One is the inclusion of steps (Figure 3.3), which are sections along the bottom of each sponson with a localized increased angle of attack. They allow the boat to get on plane more easily since the running surface can have a high angle of attack without the overall angle of attack of the boat needing to be as high. The design also has a high deadrise angle. This is the angle the bottom of the sponsons makes with the horizontal (Figure 3.4b). Having a high deadrise angle improves handling in rough conditions and during high-speed turns. The C5000 also has several lifting strakes, which also function as spray deflectors. These both help the boat get on plane faster, and deflect spray downwards, reducing spray-related drag.



Figure 3.1: Mystic C5000, Reproduced from [3]

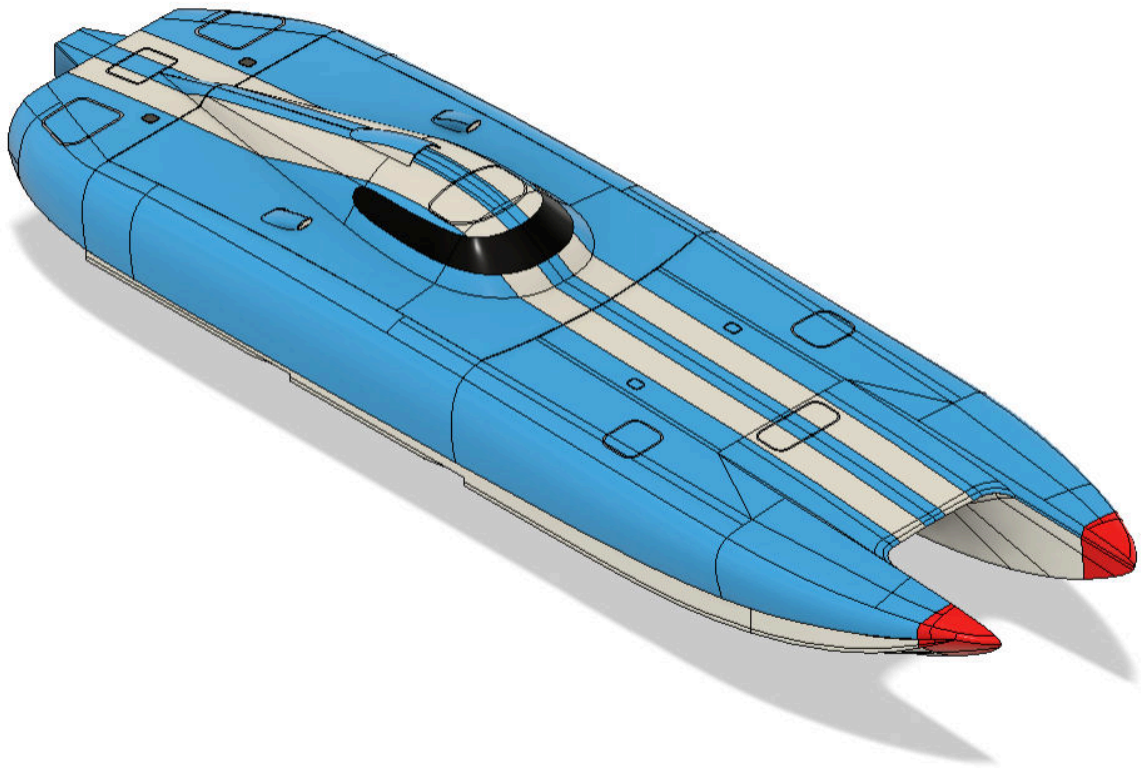


Figure 3.2: Initial Mystic C5000 CAD Model [4]

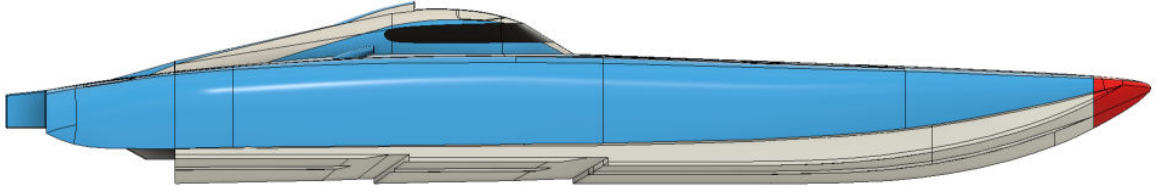


Figure 3.3: Mystic C5000 CAD Side View

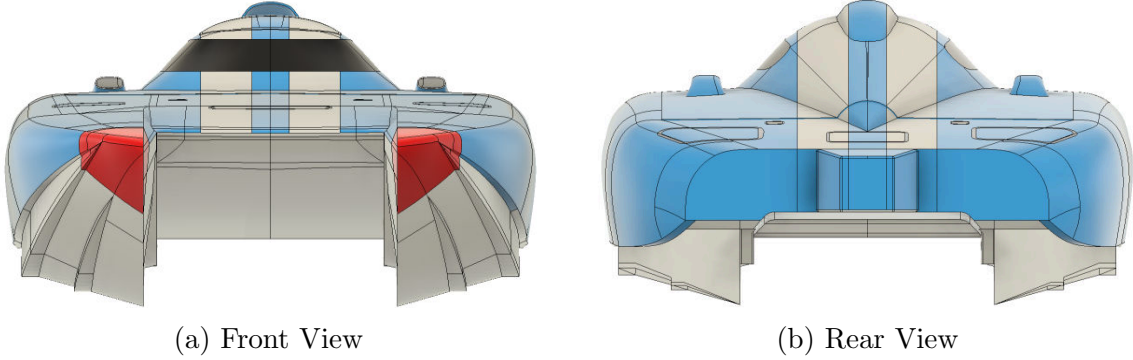


Figure 3.4: Mystic C5000 CAD Model Views

Using this model as a starting point, we determined an optimal scaling for our design requirements.

3.1.2 Optimized Scaling

In order to find the optimal scaling for the hull, we used the formulation described in James D. Russell's *Secrets of Tunnel Boat Design* [5]. The book discusses the many advantages of tunnel hulls and important design considerations. The procedure lays out initial inputs, such as weight, length, and tunnel dimensions, and determines the required power for the boat to achieve a given max speed. For this project, this procedure was modified to account for our predetermined powertrain and corresponding power output. Considering the available power we had, the hull sizing was chosen to make the appropriate amount of aerodynamic lift for our estimated weight, as well as to maximize top speed. We iterated through the calculations several times with different starting parameters until we had found the best utilization of our power output. Overall length and tunnel width were the two primary dimensions that were independently modified for this process. The entire boat was scaled down proportionally to the overall length of the boat. Then, the width of the tunnel was adjusted for that scale.

Aerodynamic Lift

The first step in the process is to determine aerodynamic lift from the tunnel, L_A . In Equation 2.1.1, ρ is air density, V is the chosen max speed, S is the reference area of the tunnel, and C_{LA} is the tunnel's coefficient of lift.

$$L_A = \frac{1}{2} \rho V^2 S C_{LA} \quad (3.1.1)$$

The coefficient of lift comes from Graph 1 (Figure 3.5) in *Secrets of Tunnel Boat Design* [5] and depends on several factors relating to the geometry of the hull.

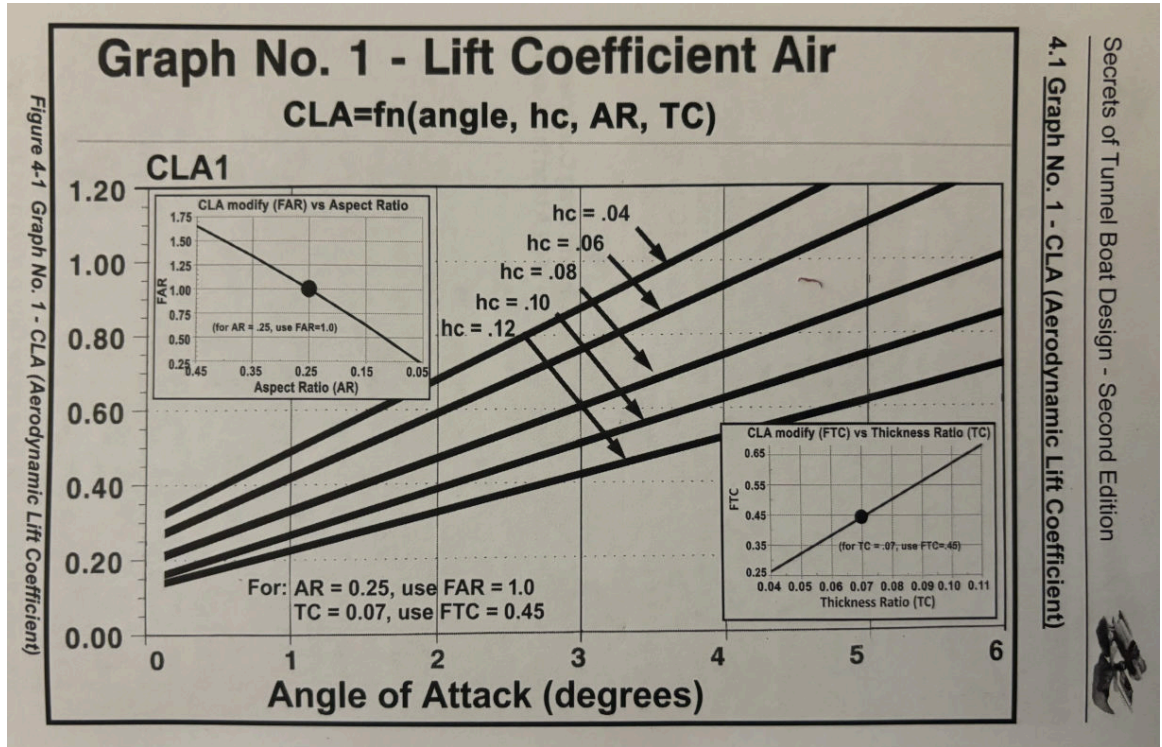


Figure 3.5: *Secrets of Tunnel Boat Design* Graph 1: Lift Coefficient Air, C_{LA} [5]

First, $CLA1$ is determined from the angle of attack of the tunnel, α_{wing} , and its height ratio, hc . hc is the ratio of the tunnel height at the half chord length to the chord length of the tunnel.

$$hc = \frac{h}{c} = \frac{H + \frac{c}{2} \cdot \sin \alpha_{wing}}{c} \quad (3.1.2)$$

H is the tunnel height in the back of the tunnel, and c is the chord length. α_{wing} is a sum of two angles: α_{trim} and α_{aero} . α_{trim} is the overall angle of attack the bottom of the hull makes with a flat running surface, while α_{aero} is the angle the tunnel chord

line makes with the bottom of the hull. A desired α_{trim} angle is provided by the designer. α_{aero} is a defined dimension of the hull; 1.8° in our case. The angle of attack used in Figure 3.5 is α_{wing} . Once $CLA1$ is determined, it must be adjusted according to the tunnels aspect ratio, AR and thickness ratio, TC . Thickness ratio, like height ratio, is simply the ratio of the tunnel's wing thickness to its chord length.

$$TC = \frac{t}{c} \quad (3.1.3)$$

The parameters used to adjust C_{LA} are FAR and FTC . They are determined based on the aspect ratio and thickness ratio of the tunnel, using the graphs in the upper left and lower right of Figure 3.5.

$$C_{LA} = CLA1 \cdot FAR \cdot FTC \quad (3.1.4)$$

Once C_{LA} is determined, L_A can be calculated using equation 3.1.1. A target for the hull design is for the boat to make 30-40% of its total weight in aerodynamic lift [5]. This is a good middle ground to aim for to minimize any risk of the boat flipping over backwards after a sudden jump in trim angle at high speeds.

Aerodynamic Drag

Next is aerodynamic drag, D_A . Aerodynamic drag depends on the same variables as lift, except it uses a reference area that's an average of the tunnel area and deck area, S_{avg} .

$$D_A = \frac{1}{2} \rho V^2 S_{avg} C_{DA} \quad (3.1.5)$$

C_{DA} comes from Graph 2 in the *Secrets of Tunnel Boat Design* [5] and is dependent on $CLA1$ and hc .

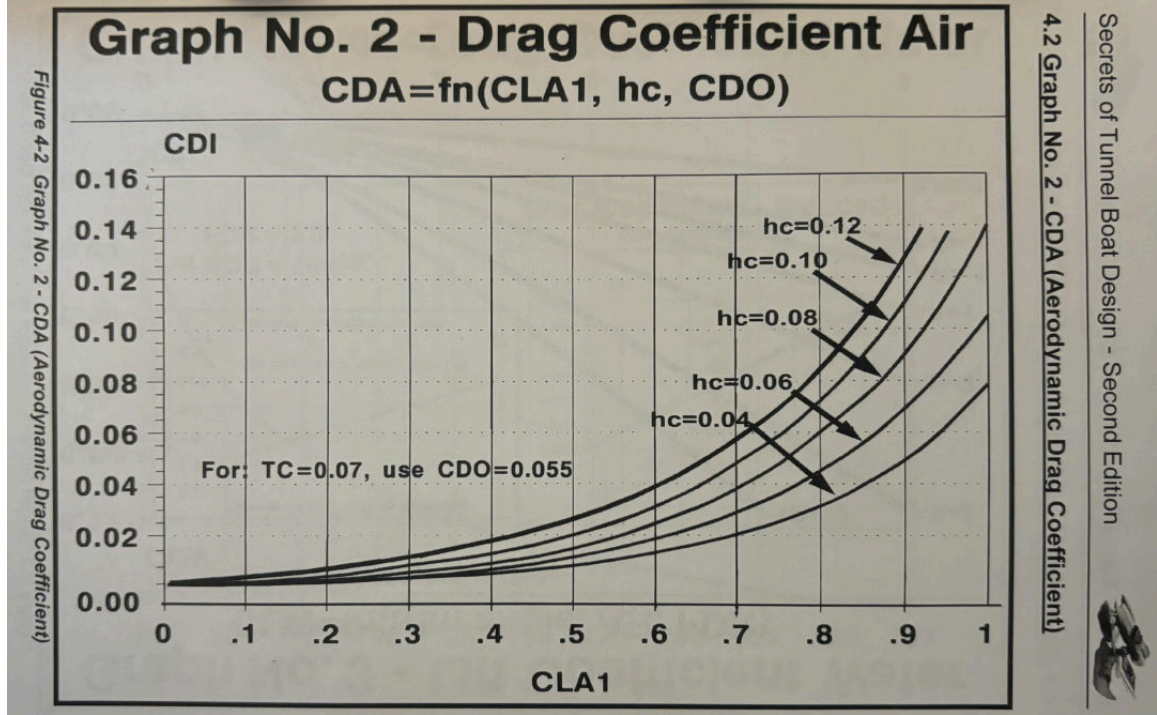


Figure 3.6: *Secrets of Tunnel Boat Design* Graph 2: Drag Coefficient Air, C_{DA} [5]

Using this graph, one can determine CDI , the coefficient of induced drag. The overall coefficient of drag, C_{DA} , is a sum of CDI and CDO , the coefficient of friction drag.

$$C_{DA} = CDI + CDO \quad (3.1.6)$$

CDO scales proportionally with TC with a reference point of when $TC = 0.07$, $CDO = 0.055$. Once C_{DA} is determined, D_A can be calculated using equation 3.1.5

Hydrodynamic Lift

After aerodynamics comes hydrodynamic lift, L_W . The hydrodynamic lift is the remaining weight of the boat not being carried by the aerodynamic lift from the tunnel.

$$L_W = W_{tot} - L_A \quad (3.1.7)$$

The amount of hydrodynamic lift required determines how much of the boat will touch the water when running. If less of the boat is in the water, then there's less hydrodynamic drag. Hence the advantage of a tunnel hull.

Wetted Area

The wetted area of the boat is the area of the sponsons touching the water while running. It is derived through an iterative process using equation 3.1.8.

$$S_W = \frac{2L_W}{\rho_W V^2 C_{LW}} \quad (3.1.8)$$

C_{LW} is determined from Graph 3 in the *Secrets of Tunnel Boat Design* [5]. It is dependent on the angle of attack of the running surface, $\alpha_{sponson}$, and the wetted aspect ratio, AR_W .

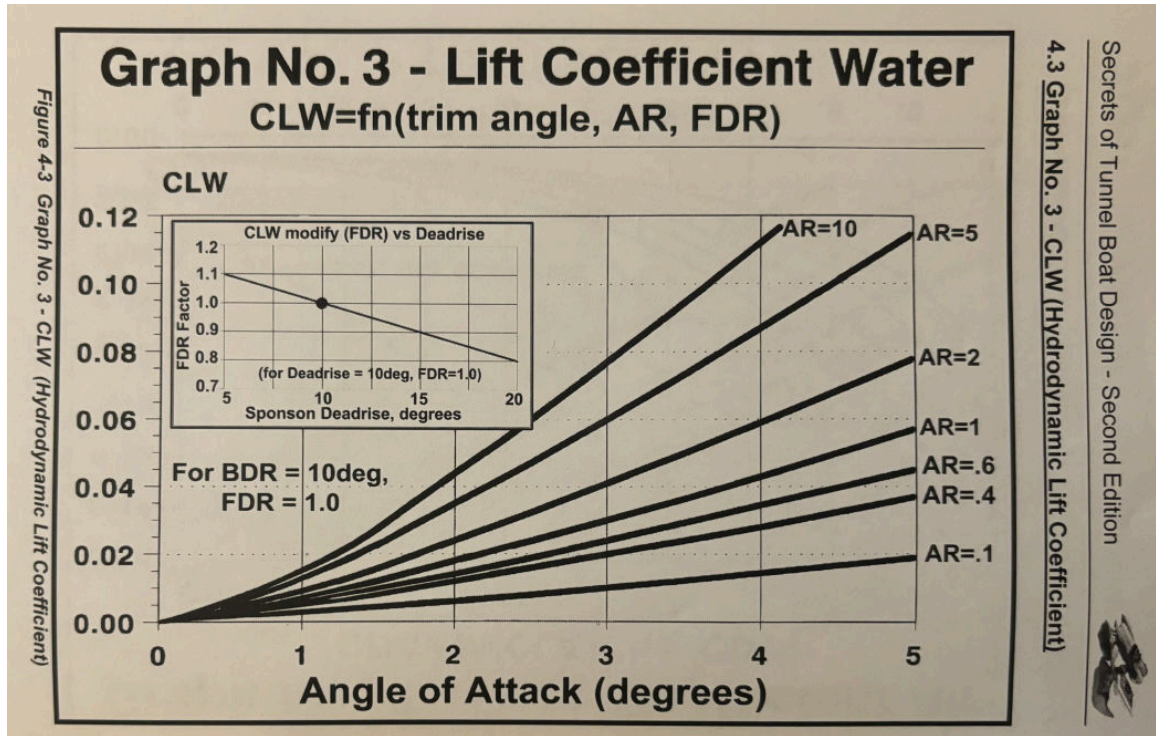


Figure 3.7: *Secrets of Tunnel Boat Design* Graph 3: Lift Coefficient Water, C_{LW} [5]

$\alpha_{sponson}$ is the sum of α_{trim} and the local angle of the rear-most step, α_{hydro} . For the C5000, this step angle is 1.45° . We have previously stated α_{trim} is chosen by the designer. However, that choice is guided by the step angle. Typically, the step angle should be about half of $\alpha_{sponson}$ for optimal performance.

Before C_{LW} can be found, $CLWb$ needs to be determined from Figure 3.7. $CLWb$ needs to be modified with respect to the deadrise angle via the FDR factor (Equation 3.1.9). FDR and deadrise angle have a linear relationship which can be seen in the upper left plot in Figure 3.7. The deadrise angle for the C5000 on the rearmost step is 21.5° .

$$C_{LW} = CLWb \cdot FDR \quad (3.1.9)$$

Now we have everything we need to calculate wetted area. The iteration process starts with the designer making an initial guess for S_W . The wetted length, l_W , and wetted aspect ratio, AR_W , are calculated using this guess, as well as the sponson width, b .

$$l_W = \frac{S_W}{2b} \quad (3.1.10)$$

$$AR_W = \frac{b}{l_W} \quad (3.1.11)$$

Using AR_W and $\alpha_{sponson}$, a new value of $CLWb$ is determined from figure 3.7. The modified C_{LW} is then calculated using Equation 3.1.9. Next Equation 3.1.8 is used to acquire a new S_W value, completing one iteration. If this S_W is approximately the same as the initial guess, then it is the correct wetted area. However, if it is not the same, then the process is repeated, starting this time with the new S_W value. Continue iterating until the output S_W value is the same as that of the prior iteration. The level of precision is up to the designer, but we iterated until the wetted area was within 0.001 ft^2 of the previous iteration.

Hydrodynamic Drag

Now that we know how much of the boat is in the water, we can determine the hydrodynamic drag, D_W , experienced by the hull.

$$D_W = \frac{1}{2} \rho_W V^2 S_W C_{DW} \quad (3.1.12)$$

Several sources of drag contribute to C_{DW} as shown in Equation 3.1.13. Each component of C_{DW} is described in Graph 4 from the *Secrets of Tunnel Boat Design* [5].

$$C_{DW} = CDF + CDI_W + CDP \quad (3.1.13)$$

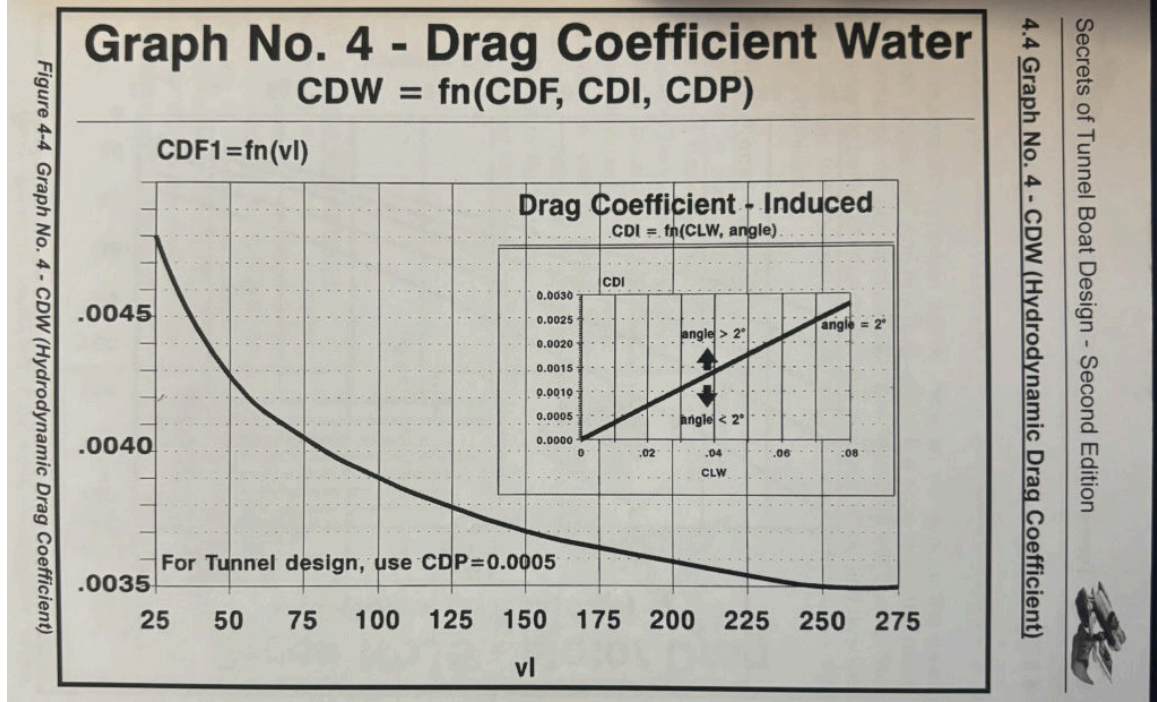


Figure 3.8: *Secrets of Tunnel Boat Design* Graph 4: Drag Coefficient Water, C_{DW} [5]

CDF is the coefficient of friction drag. It is a function of $CDF1$ and SAF .

$$CDF = CDF1 \cdot (1 + SAF) \quad (3.1.14)$$

$CDF1$ is determined from Figure 3.8 and depends on velocity length, vl . Velocity length is defined as the product of the wetted length of the sponson, l_w , and the top speed of the boat, V .

$$vl = V \cdot l_w \quad (3.1.15)$$

SAF is the spray area factor, representing the portion of the hull experiencing additional drag due to spray. A typical SAF for a tunnel hull is 0.5. Although the C5000's spray deflectors may reduce this value, 0.5 was used to conservatively overestimate drag.

CDI_w is the coefficient of induced drag and depends on C_{LW} and $\alpha_{sponson}$, determined from the plot in the upper right of Figure 3.8.

CDP is the profile drag coefficient, typically 0.0005 for tunnel hulls. With these components, C_{DW} can be calculated using Equation 3.1.13, and Equation 3.1.12 is then used to determine the total hydrodynamic drag, D_W .

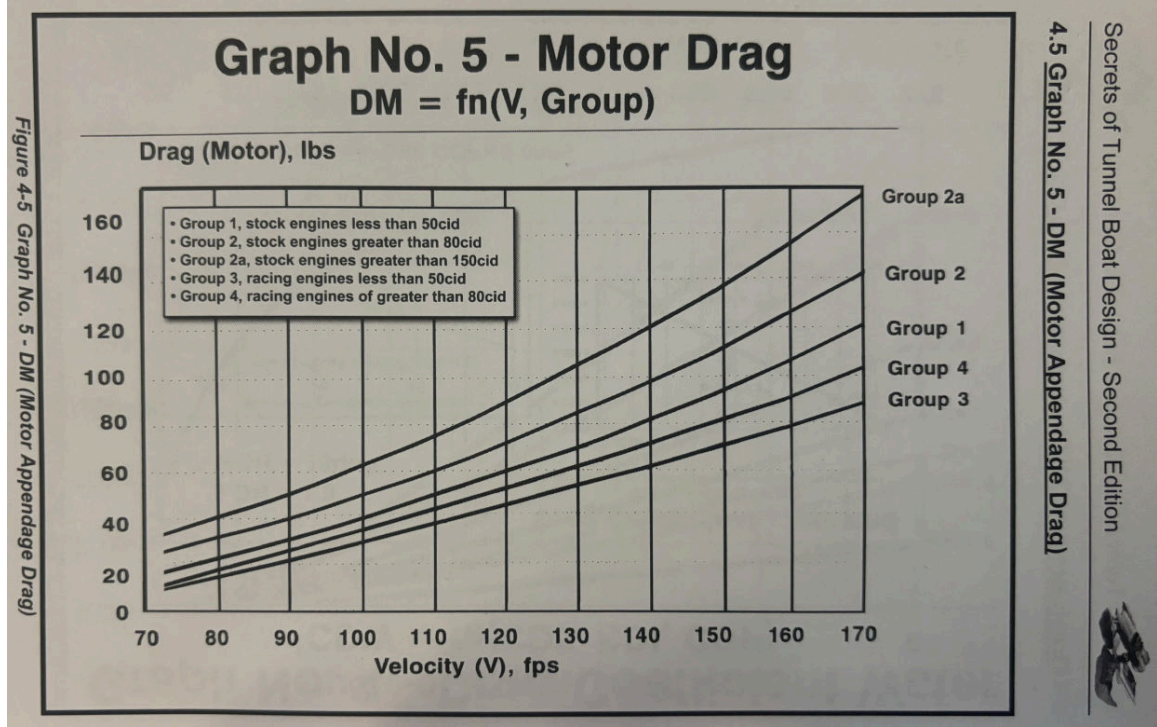


Figure 3.9: *Secrets of Tunnel Boat Design* Graph 5: Motor Drag, D_M [5]

Appendage Drag

Appendage drag is made up of any extra sources of drag not previously accounted for. This is primarily due to the propellers, rudder, and cockpit. Unfortunately, this is the one part of the process that does not scale well with RC boats. The book focuses on outboard motors and gives drag curves for various gas engine sizes in Graph 5 [5].

In talking with the author, James Russell, we decided that taking one 5th of the motor drag from Group 3 at our top speed was the best estimate for our driveline drag, D_M (Figure 3.9). This was intentionally left as an overestimate, once again to prevent overestimating the capabilities of the boat. Since our boat is inboard, and the driveline is generally much smaller, the drag would be much smaller than that of a manned tunnel boat. We then also measured the cross-sectional area of our cockpit in Fusion 360, S_A , and used it to calculate the aerodynamic drag on the cockpit. $C_{D,cockpit}$ was chosen to be a standard 0.3 for streamline closed canopy cockpits like ours.

$$D_{cockpit} = \frac{1}{2} \rho V^2 S_A C_{D,cockpit} \quad (3.1.16)$$

Power Requirement

Now every source of drag is accounted for, and the total drag experienced by the boat can be calculated.

$$D_{tot} = D_A + D_W + D_M + D_{cockpit} \quad (3.1.17)$$

Then the power required for the boat to operate under the given parameters can be calculated using Equation 3.1.18

$$P_{req} = \frac{D_{tot}V}{550} \quad (3.1.18)$$

Iterating

All of the calculations described above were automated in a MATLAB script (A.1), allowing for quick design iterations. After each iteration, several data points were compared, the most important being the available power, P_{avail} , vs required power, P_{req} . Previously, it was determined that a conservative estimate for power output at the motors was 12 kW. After accounting for a 10% loss at the shaft and 20% loss at the prop, the system is making 8.64 kW or 11.6 HP. If the power required calculated by a given iteration was lower than this, we would increase the top speed and go through the process again. If it was higher, we would either lower the top speed or widen the tunnel to make more lift. However, when widening the tunnel, we also kept watch on the total amount of lift being made, trying to steer towards the goal of 30-40% of the boat's total weight. If the hull was producing too much lift, we thinned the tunnel. If too little, we either widened the tunnel or increased top speed. The challenge came from finding a balance between maximizing top speed, not making too much lift, and not being underpowered. We iterated through the calculations many times until we found a scaling that satisfied the required range of lift while also pulling as much speed as possible out of our available power.

Final Scaling

Table 3.1 shows the dimensions we landed on for the final scaling of the hull.

Parameter	Symbol	Value	Unit
Initial Assumptions			
Weight Estimate Used	W_{tot}	130	lb
Top Design Speed	V	80	mph
Shaft Power Loss	η_s	10	%
Prop Slip Loss	η_p	20	%
Fixed / Reference Geometry			
Tunnel Angle	α_{wing}	1.8	deg
Step Angle	α_{hydro}	1.5	deg
Deadrise Angle	β	21.5	deg
Height Ratio	hc	0.0974	–
Thickness Ratio	TC	0.0877	–
Designer-Selected Variables			
Length	L	7.0	ft
Tunnel Width	w_t	18.0	in
Trim Angle	α_{trim}	1.5	deg

Table 3.1: Initial Assumptions, Fixed Geometry, and Designer-Selected Variables for the Final Hull Design

The weight estimate is based on the estimate tabulated in Table 3.2 but with an extra 10 lbs as a margin of error. The *Autonomous System* listed in Table 3.2 is a control system that was added to the boat for the autonomous division of the Promoting Electric Propulsion competition, which is outside the scope of this thesis.

Component	Weight (lb)
Hull	
Hull	40
Powertrain	
Battery	18
Motors	11
ESCs	2
Cabling, contactor, etc.	8
Rudder, props, driveshafts	2
Coolant System	
Hosing	1
Water jacket	2
Control System	
Servo	0.1875
Transmitter & antenna	0.3
Autonomous System	5
Payload	
Steel Blocks	30
Total Estimated Weight	119.4875

Table 3.2: Estimated Weight Breakdown of the Boat by Subsystem

This weight estimate is heavier than a boat of our size would normally be because of the payload. To handle the extra weight, the tunnel width was increased by 50%. A 1:1 scaling of the boat to 7 ft results in a 12 inch tunnel width. We widened the tunnel to 18 inches to give us the results in Table 3.3. With this scaling, the hull made 36% of its weight in aerodynamic lift, which fits in the ideal range. It also left us with less than 0.5 HP in overhead compared to the power required. The overall length of the hull was limited to 7 ft primarily for the convenience of being able to carry it around the shop. The length had less calculated impact on performance than the tunnel width.

Parameter	Symbol	Value	Unit
Derived Geometry			
Chord Length	c	6.04	ft
Overall Width	b_{deck}	30.5	in
Sponson Width	b	5.56	in
Tunnel Aspect Ratio	AR	0.2482	–
Calculated Performance			
Aero Lift	L_A	46.5	lb
Aero Drag	D_A	18.7	lb
Hydro Lift	L_W	83.5	lb
Hydro Drag	D_W	24.3	lb
Motor Drag	D_M	8.4	lb
Appendage Drag	$D_{cockpit}$	0.9	lb
Total Drag	D_{tot}	52.3	lb
Power Required	P_{req}	11.154	Hp
Power Available	P_{avail}	11.5	Hp

Table 3.3: Derived Geometry and Predicted Performance for the Final Hull Design

Figures 3.10 through 3.12 show the final scaling of the hull in CAD. The cockpit was expanded with the tunnel for ease of design, but it makes minimal difference in performance. The air traps were also removed because this is an electric boat. Finally, the rudder block off the back of the transom was removed in order to make way for a custom 3D-printed rudder mount. The rudder block in the CAD would have likely been too big for any rudder we could have found and would have been hard to manufacture properly.

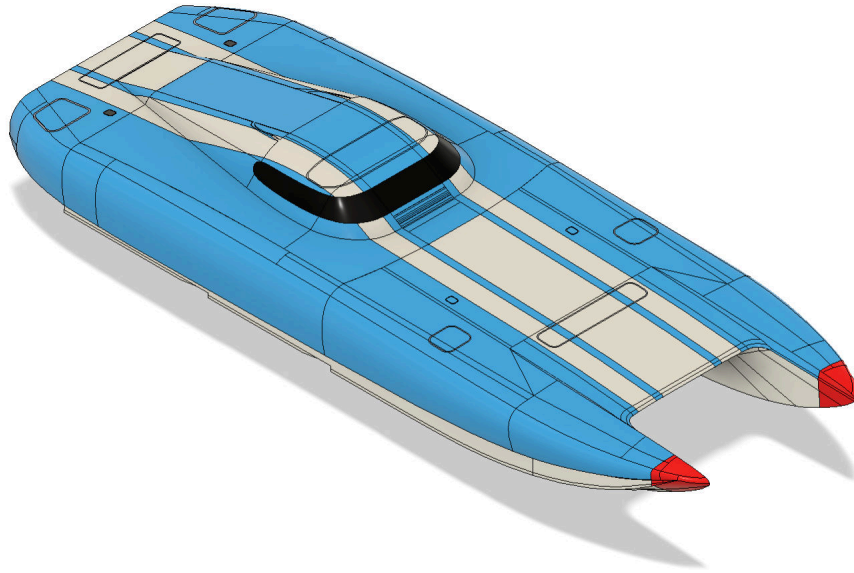


Figure 3.10: Final Scaling CAD

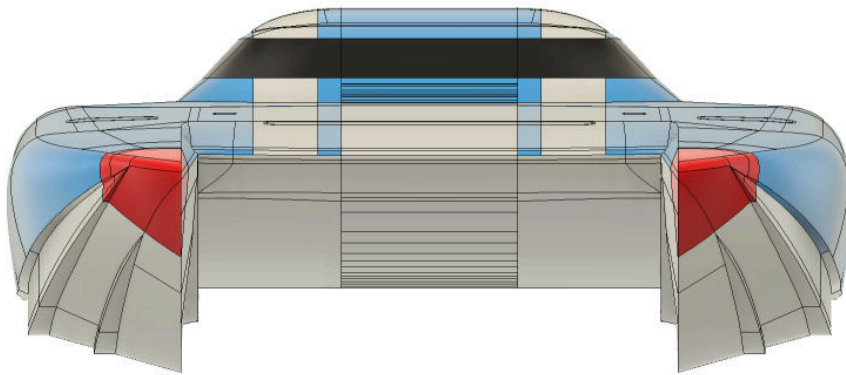


Figure 3.11: Final Scaling Front View

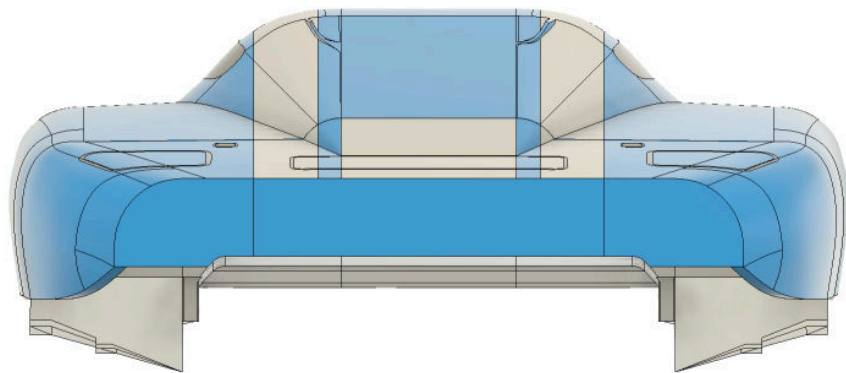


Figure 3.12: Final Scaling Rear View

3.2 Powertrain

The goal in selecting a motor and ESC was to find a setup under 55V that had the best power density, a reasonable cost, and was easy to use with data logging capabilities. After searching through multiple options, we settled on the Castle 2535 Sensored Motor - 680Kv paired with the Hydra Cobra 5 HV, 12S, 50.4V ESC [6] [7]. This setup features high power density with the capability of taking 180A continuous at 50.4V, outputting 9 kW of power. Instead, we are running 44.4V at around 140A for 6 kW. With a twin motor configuration, that provides 12 kW, at 120 lbs of displacement, the theoretical top speed is 80 MPH. If implemented properly and with a successful autonomous and electrical system, this setup would provide enough power to succeed in the PEP competition.

The Turnigy High Capacity 6S 16000mAh 22.2V 12C Lipo Battery was selected to run in a 2S configuration per ESC to achieve 44.4 V nominal.



(a) Castle 2535 Sensored Motor - 680Kv
[6]



(b) Hydra Cobra 5 HV, 12S, 50.4V ESC
[7]



(c) Turnigy High Capacity 6S 16000mAh
22.2V 12C Lipo Battery [8]

Figure 3.13: Motor, ESC, and Battery Selection

3.3 Driveline

The driveline consists of components that transfer power from the motor to the propellers to generate thrust. Starting from the motor shaft, it includes a thrust bearing, a shaft clutch collet, a flex shaft, a brass guide tube, a stinger, a propeller shaft, and a propeller.



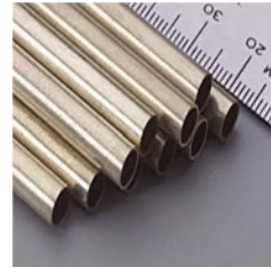
(a) Thrust Bearing [9]



(b) Shaft Clutch Collet [10]



(c) Flex Shaft [11]



(d) Brass Stuffing Tube [12]



(e) Stingers with Propeller Shafts [13]



(f) Propeller [14]

Figure 3.14: Driveline Components

On one end, the shaft clutch collet is set screwed onto the motor shaft, and on the other end, the flex shaft is evenly clamped down into the collet using a nut, allowing for easy installation, even clamping force, and concentricity with the motor. A thrust bearing slides around the motor shaft and sits in a circular indentation in the motor mounting plate, and the collet sits against it to allow the plate to partially absorb thrust loads through the plate instead of the internal motor bearings. A “stuffing

tube” is a brass tube used to guide the flex shaft from the motor and into the stinger. The stuffing tube end is epoxied to the inside of the sponson rear wall and supported with a guide plate. A stinger is a shaft mounted to both the sponson rear wall and the transom overhang wall that guides the flex shaft and holds the propeller in place. The flex shaft is fixed to the propeller shaft using Loctite 638, a strong binding agent for close-fitting metal cylindrical parts [15] [16].

There are several common driveshaft options: flex shaft, wire drive, and solid shaft. Flex shafts are multistrand cables wound together, which allows for misalignments, but they also lose power transfer efficiency due to their lower stiffness. Solid shafts are the opposite because they are made of solid steel, which provides the greatest efficiency but also requires much more precise alignment of the motor and stinger. Wire drive is the in-between, being a single solid spring steel wire that requires tensioning. The flex shaft was ultimately selected for its successful heritage, accessibility, and ability to absorb small misalignments that would likely exist.

To accommodate the flex shaft, each motor was set at a 9.5° downward angle, with the stuffing tube bent to continually guide the flex shaft from the motor to the propeller shaft. This gives the flex shaft a wide, continual bend that transitions to the stinger. A bend is necessary to prevent the shaft from flailing at higher RPMs. A minimal single bend was achieved while considering the size of the motor and water jacket to fit in the sponson [17]. This bend was achieved using a bending jig designed and 3D-printed, and shown in Figure 3.15. The end of a straight tube is placed in the notch on the right, and the curved tool is pressed into the slot in the jig to curve the tube.

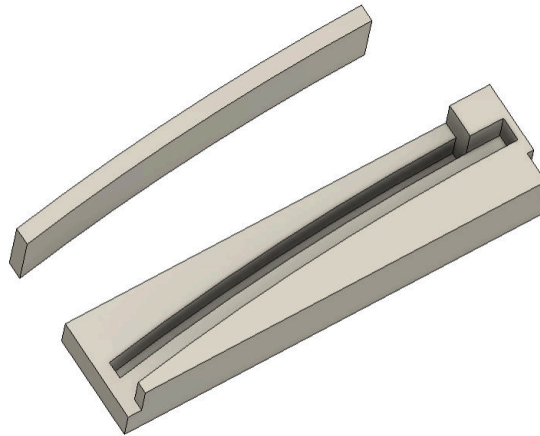


Figure 3.15: Pipe Bending Jig

There are four plates throughout the driveline: the middle motor support, the front motor support, and two stuffing tube supports. The middle motor support holds some of the weight of the motor to relieve the bolts and front motor support of stress from the motor's weight. The front motor support has the motor bolt hole pattern to mount the motor on, the shaft hole, and the thrust bearing indent. Finally, the two tube supports are used to locate the stuffing tube.

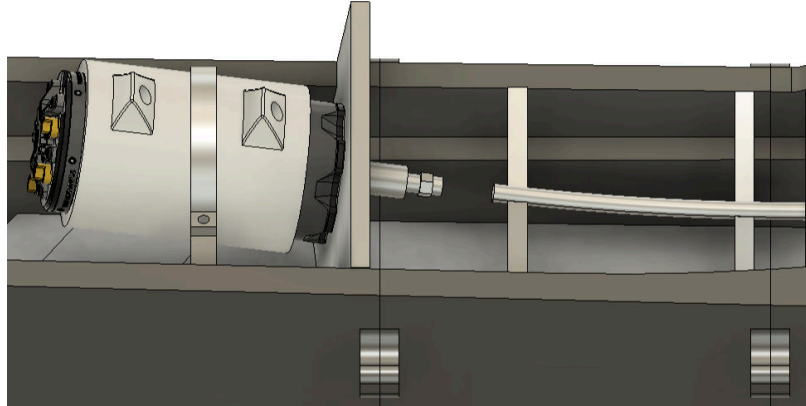
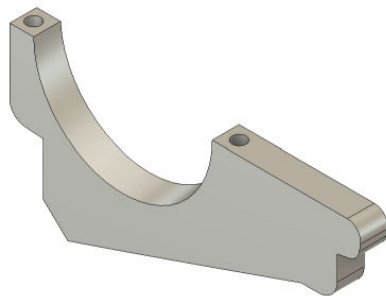
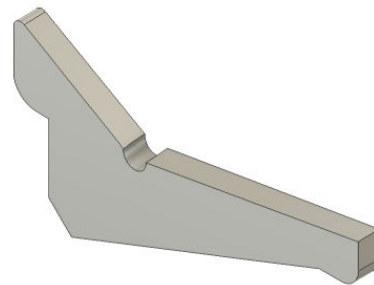


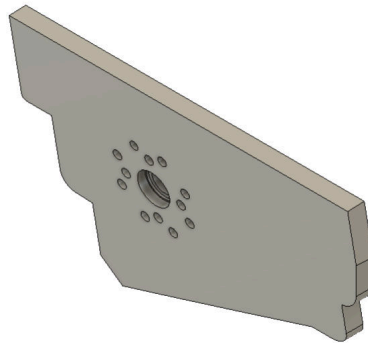
Figure 3.16: Driveline Plates Installed in Sponson



(a) Middle Motor Support Plate



(b) Tube Support Plate



(c) Front Motor Support Plate

Figure 3.17: Driveline Plates

3.4 Steering

3.4.1 Component Selection

A small-scale catamaran that runs a twin inboard motor setup requires steering through either differential steering or rudder steering. The traditional single, centered rudder was implemented in case differential steering could not be achieved for on-water testing. The rudder is directed by a servo with a cable tied to each end of its horn. This cable is routed through an actuator that attaches to the rudder. The MHZ Mystic C5000 rudder, designed for a 1850-1890 mm model, was selected for its closest compatibility with Osprey because it is designed for the same catamaran model of a similar scale (Figure 3.21).

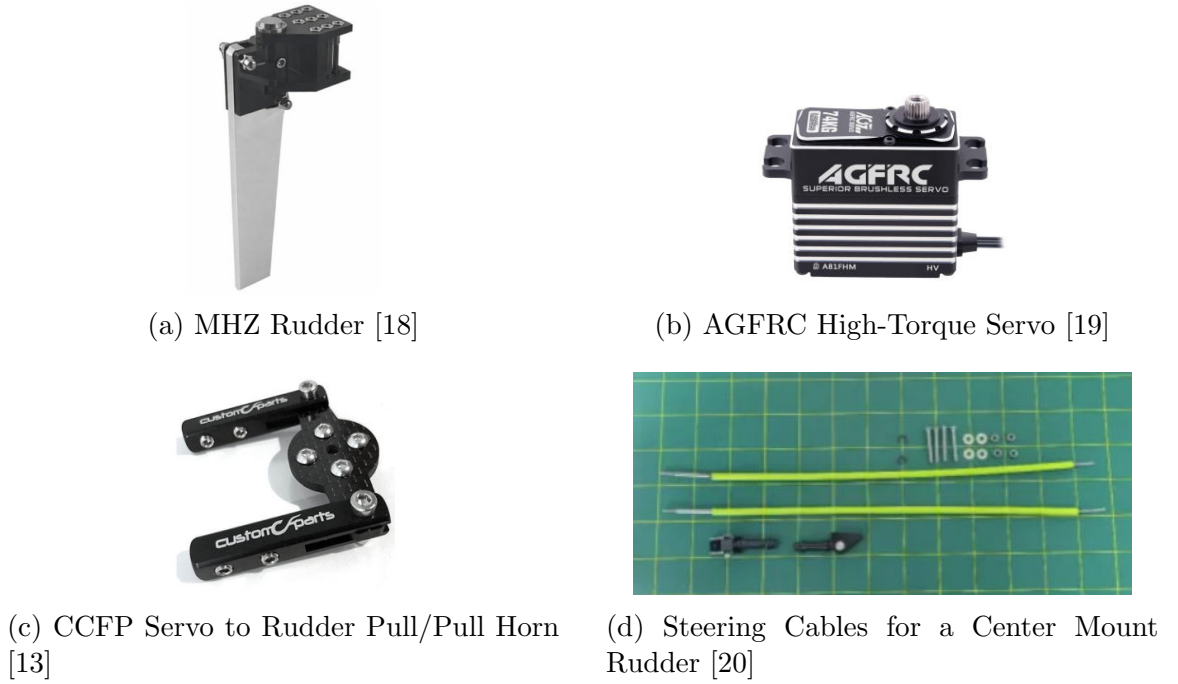


Figure 3.18: Steering Components

With rudder dimensions known, the maximum required torque from a steering servo can be calculated. The force is dominated by the square of the vessel velocity, so the maximum speed condition is considered. Drag force, shown in Equation 3.4.1 [21], is calculated using an expected vessel speed of 80 mph, brackish water density of 1010 kg/m^3 , submerged blade area of 50% of 150 mm by 30 mm, and estimated coefficient of drag of 0.3 [22].

$$F_{\text{Drag}} = \frac{1}{2} \rho_w V^2 A_{\text{rudder}} C_{D,\text{rudder}} \quad (3.4.1)$$

The force comes out to be 436 N, which, with a servo horn moment arm of 1.5 cm, requires 66 kg·cm of torque. The AGFRC A81FHM HV 74KG Super Torque 4-Poles Brushless Motor Programmable Digital Servo [19] was selected, providing a stall torque of 74 kg·cm at 8.4 V. The voltage is provided by one of the ESCs via a Battery Elimination Circuit (BEC), which provides a step-down voltage of up to 8.5 V from the high-voltage batteries. The servo is paired with the CCFP Servo to Rudder Pull/Pull Horn shown in Figure 3.18c [13], and cables shown in Figure 3.18d are screwed onto the horn arms and bolted to the rudder arms.

3.4.2 Rigging Design

Servo Mount

The goal with the servo mount was to make it structural while placing the servo as low as possible. The servo was mounted as far forward as possible on the elevated section in the back of the tunnel. The mount was designed to be a two-piece 3D printed assembly. The servo slides into one side of the mount before being bolted down. Then the second piece slides in behind it and is pinned into place. This design allows the servo to be fully encased in the mount while still leaving room for the wires to be routed through.

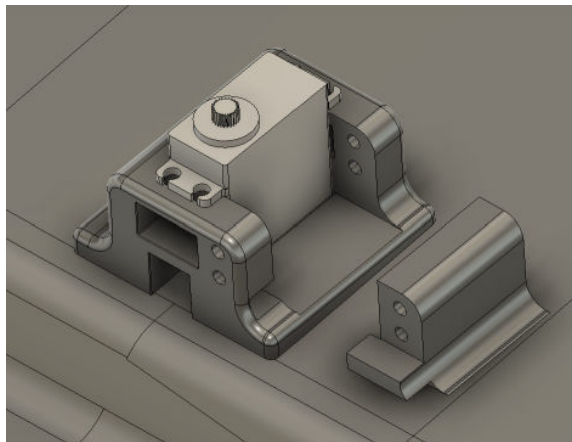


Figure 3.19: Servo Mount

Steering Cable Routers

The steering cables are routed through teflon tubes from the servo arms to the rudder. A routing mount was designed to hold one end of each tube up to the servo horn.

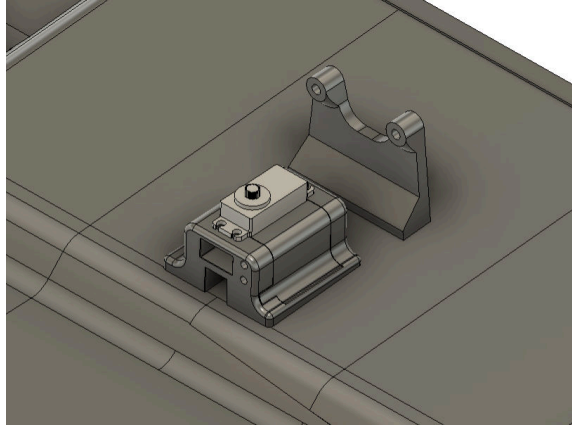


Figure 3.20: Steering Cable Router

Rudder Mount

The primary criterion for the rudder mount was mounting the rudder as low as possible. There were concerns that the rudder we had was not long enough to get enough engagement with the water. It could not be mounted lower than the bottom of the transom because, at high speeds, it would cause too much drag, and at low speeds, the mount would be dragging through the water. Figure 3.21 shows the rudder mount design with tabs for the steering cable actuators to mount to.

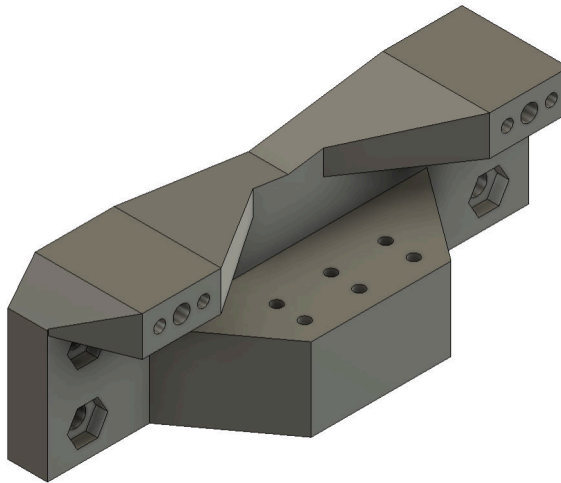


Figure 3.21: Rudder Mount

A challenge that arose with this low rudder mount was the difference in height between the steering cables and the arms on the rudder. To compensate, a rudder block was designed to mount atop the rudder and connect to the steering cables.

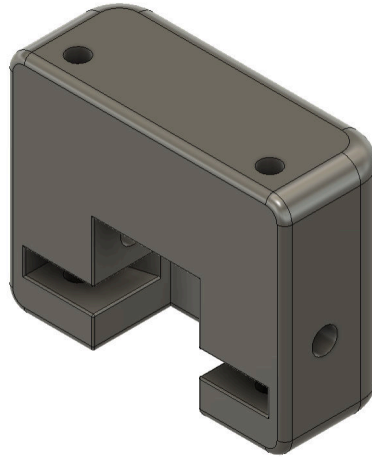


Figure 3.22: Rudder Block

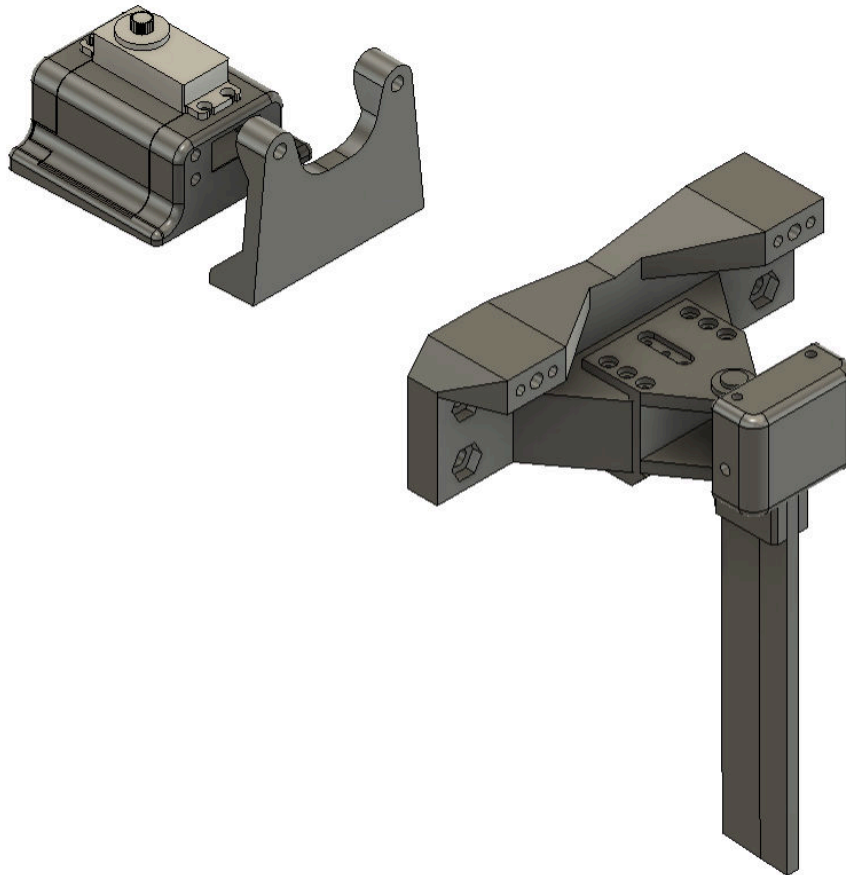


Figure 3.23: Steering Assembly CAD

3.5 Coolant System

Cooling is necessary to maintain safe component operating temperatures when demanding high power from the motors and ESCs. The major decision before designing the system was whether to run raw water or deionized water through the components. The PEP competition is held at Portsmouth City Park, VA, in the brackish Elizabeth River. Running brackish water through the vessel is a significant concern for electrical shorting and corrosion. However, running a separate deionized water line increases the number of components. Ultimately, deionized water was selected to avoid saltwater on critical components. The selected ESCs have built-in water cooling, a necessity when pulling 100A+ of current at 44 V at high speeds. The motors do not, but they have an exterior profile that an outer water-cooled jacket can slide onto. Because there are no existing commercial water jackets designed for the size of motor selected, a custom water jacket was designed and printed (Figure 3.24a). It closely follows typical existing aluminum jackets, with two hose fittings and an enclosed cylindrical volume to contain flowing water sealed on both ends by radial O-rings [23]. This design allows a single-body 3D print to be slid onto the motor can.

The cooling setup has one closed deionized water loop and one raw water pickup line per motor-ESC pair, sharing a common water reservoir. Each closed loop starts at the reservoir and is pumped using an electric pump through an inline filter, ESC, water jacket, heat exchanger, and back to the reservoir. The reservoir and water jackets are designed and printed using ASA (Acrylonitrile Styrene Acrylate) filament from Elegoo, a material known for its temperature, UV, and general outdoor exposure resistance [24]. The raw water pickup tube is through the heat exchanger and escapes at an outlet.

Maintaining the motors and inverters within their operational temperature limits is essential to system reliability and long-term performance. The cooling system must satisfy this thermal requirement. To meet these requirements, the cooling system was designed around several key functional objectives:

- **Heat Loss Characterization:** The thermal output of the motors and ESCs must be quantified to determine the total system heat load.
- **Heat Exchanger Selection:** A suitable heat exchanger must offer sufficient thermal contact area between the coolant and the external environment to enable efficient heat rejection.
- **Pump Selection:** A suitable pump must offer sufficient flow rate to enable

efficient heat exchange.

Documentation does not list the efficiency of the motors and ESCs. We will assume a motor efficiency of 90% and ESC efficiency of 90% at full throttle [25]. With 6kW of power draw, these inefficiencies would result in heat loss, Q , of 1.14kW according to Equation 3.5.1 where P is input power and η is total efficiency.

$$Q = P \cdot (1 - \eta) \quad (3.5.1)$$

The two main components to select are the pump and the heat exchanger. They dictate water flow rate and heat-exchanging surface area. The CrocSee DC 12 V 1.3 LPM water pump [26] through an inline filter [27] was selected for usability with deionized water, low cost, small volume, and compatibility with a 12 V battery. The following calculations were done to verify that the B3-5A 10 Plate Heat Exchanger, featuring two fluid lines that exchange heat via counter-flow, is sufficient for the cooling system [28].

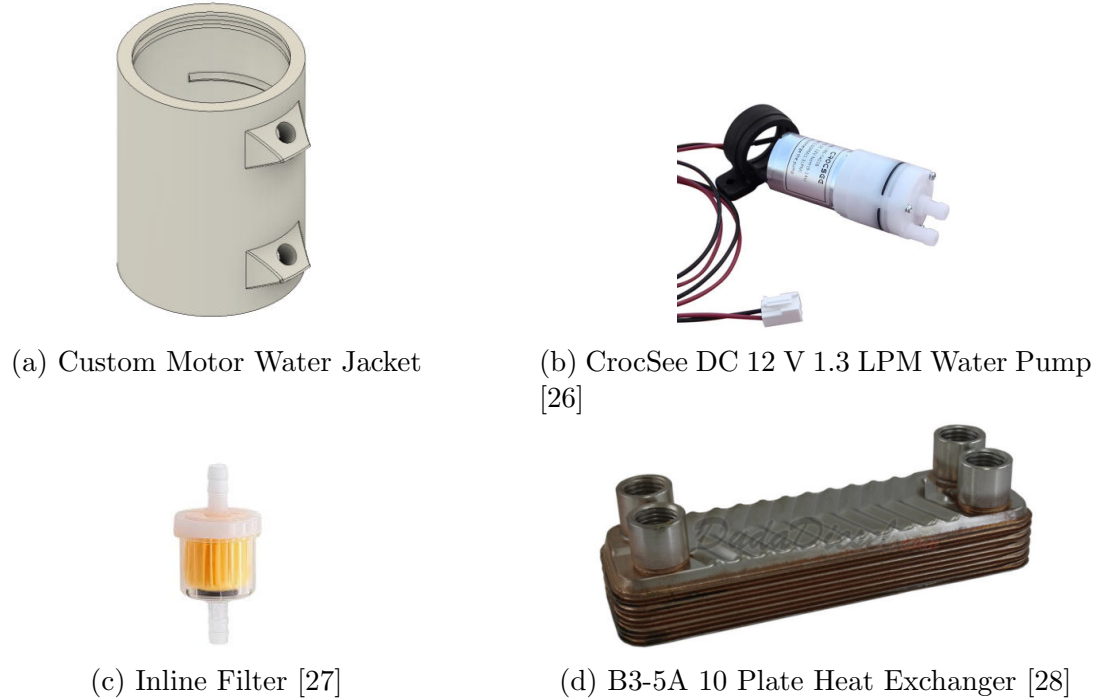


Figure 3.24: Cooling System Components

Equation 3.5.2 is the steady state equation for convective heat transfer. The external raw water temperature can be estimated at 20°C. $\dot{m}_2$ is the mass flow rate of the water pickup and c_p is the specific heat capacity of the raw water.

$$Q = \dot{m}_2 \cdot c_p \cdot (T_{out} - T_{in}) \quad (3.5.2)$$

The mass flow rate of raw water is dependent on the vessel speed and the cross-sectional area of the pickup tube. It is calculated using Equation 3.5.3. Assuming 40 MPH speed, brackish water density of 1010 kg/m^3 , and pickup tube diameter of 5 mm, the flow rate is 0.35 kg/s .

$$\dot{m}_2 = v \cdot \rho_W \cdot A_{tube} \quad (3.5.3)$$

Then, using Equation 3.5.2 and $Q=1.14 \text{ kW}$, $c_p=4 \text{ kJ/(kgK)}$, and $T_{in,2} = 20^\circ\text{C}$, $T_{out,2}$ is calculated to be 20.81°C .

The Log Mean Temperature Difference (LMTD) method is used for this calculation, shown in Equation 3.5.4 [29].

$$LMTD = \frac{(T_{in,2} - T_{out,1}) - (T_{out,2} - T_{in,1})}{\ln\left(\frac{T_{out,2} - T_{in,1}}{T_{in,2} - T_{out,1}}\right)} \quad (3.5.4)$$

$$A = \frac{Q}{(\alpha \cdot LMTD)} \quad (3.5.5)$$

Using Equation 3.5.5, the LMTD is calculated using the heat exchanger surface area, A , of 0.05 m^2 , an estimated heat transfer coefficient, α , of $3000 \text{ W/(m}^2 \cdot \text{K)}$ [30], and $Q = 1.14 \text{ kW}$. LMTD comes out to 7.6°C .

Knowing $T_{in,1}$, $T_{out,2}$, and assuming $T_{in,1} = 40^\circ\text{C}$, $T_{in,2}$ can be calculated as 22°C , a significant decrease from 40°C and a safe operating temperature. Even with motor and ESC temperatures beyond 40°C , temperatures can be lowered to well within the maximum allowable temperature of 82°C .

3.6 Component Mounting

3.6.1 Battery Rails

The desired CG of the vessel lies at around 30% of the length of the hull from the transom [31][32]. However, the desired CG ultimately depends on performance characteristics of the individual hull, environmental conditions, and target speeds. Battery packs and the required 30 lb payload are the two heaviest components in the vessel, making them the best components whose locations can be adjusted. A sliding rail design was devised, shown in Figure 3.25. Flat trays were designed for the batteries and

payload. The battery mounts have slots to insert velcro straps, allowing for quick removal, yet strong fastening. Payload mounts use bolted clamps with embedded nuts. Embedded nuts are fully encased in a 3D printed part and act as an inserted thread. Each tray is held down to the rails using four bolts whose hex heads each press fit into a sliding block, shown in Figure 3.26.

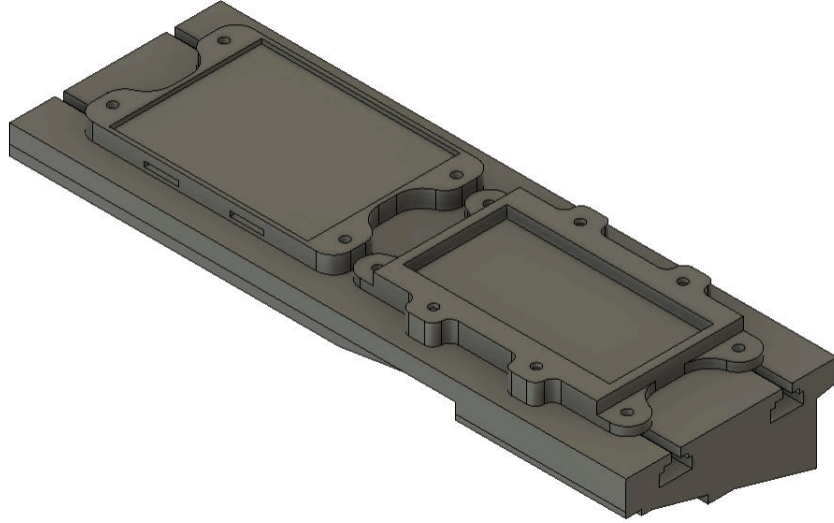


Figure 3.25: Battery rails with battery and payload mounts

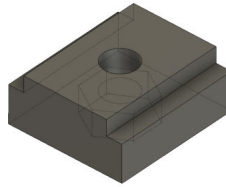
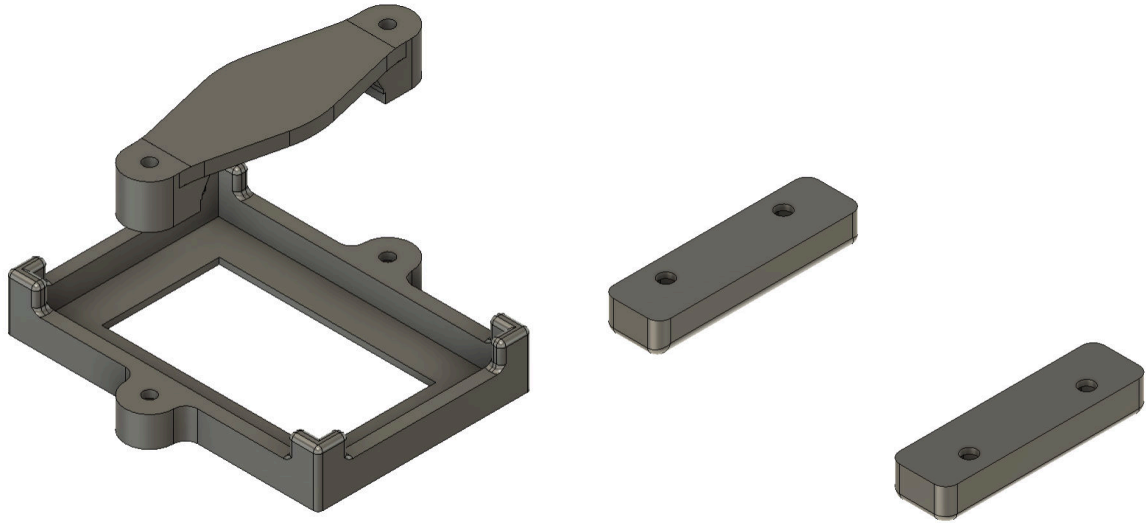


Figure 3.26: Battery Rail Slider Nut

3.6.2 General Component Mounting

The mounting for most general components fell into one of two categories: direct bolting to a 3D print or clamped to 3D print. If a component had accessible bolt holes on it, such as the kill switch box, then it bolted directly to a mount which would be attached to the hull. If not, such as with the ESCs, a print was designed to bolt down to the mount, clamping the component down with it. Both methods utilized embedded nuts. Examples of both styles of mount are displayed in Figure 3.27.



(a) LV battery mount (clamp-style)

(b) Kill switch mount (direct bolt)

Figure 3.27: Mounting styles for general component mounting

3.7 Electrical System

The electrical architecture is organized into two independent voltage domains: a 44.4 V nominal high-voltage (HV) propulsion system and a 12.8 V low-voltage (LV) auxiliary and control system. This separation ensures that a fault on the propulsion side cannot collapse the logic and safety systems, and that the remote kill switch remains operational during any failure.

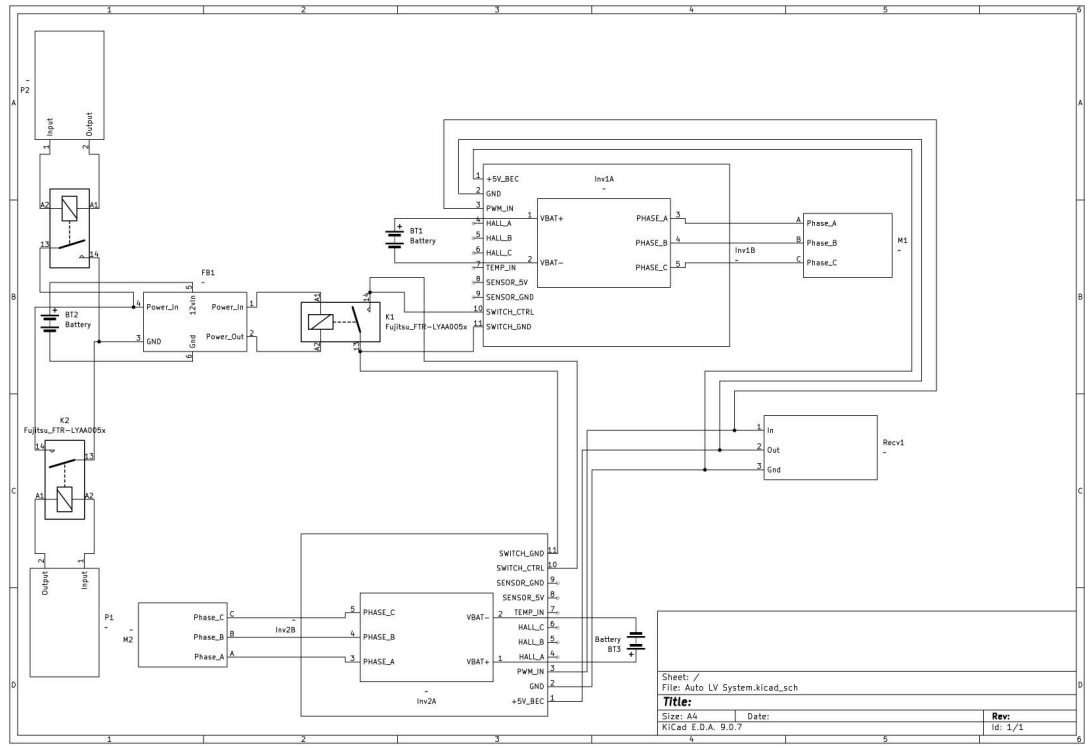


Figure 3.28: KiCad schematic of the LV electrical system

3.7.1 High-Voltage Propulsion System

The HV system consists of two fully independent drive setups. Each includes a 44.4 V battery pack (two 22.2 V LiPo packs in series), an ESC, and a motor. The channels share no common HV rail; each battery, ESC, and motor set is electrically isolated from the others. This means a cell failure, ESC fault, or phase-short in one cannot directly propagate to the other. Each HV connector in the system was replaced with QS8 antispark connectors.

The ignition system is the interface between the LV control domain and the HV propulsion domain. Each ESC has a built-in ignition switch. This switch has been rewired to connect to a remote relay that will receive a signal from an operator on land. This relay has a roughly five-mile radius to ensure safety. If the ignition switch is turned off while the boat is in operation, the inverters immediately cut out and stop propulsion.

3.7.2 Low-Voltage Auxiliary and Control System

The GOLDENMATE 12V 10Ah LiFePO4 Lithium Battery was chosen to provide the low voltage power supply due to it having an internal battery management system that protects it against shorts [33]. The LV system governs all relay switching, logic, water pump cooling, and RC receiver communication. The 10 Ah capacity is substantially oversized relative to the draw of the relay coils, RC receiver, and pump motors, providing longer run times.

Relays perform the critical switching and gating functions in the LV system. Coolant pumps are controlled by switch-operated relays, and the ESC ignition is governed by a remote dry relay.

A six-position blade fuse block distributes 12.8 V power independently to each LV load. Individual fusing means a fault in any single circuit, such as a stalled pump motor or a wire short, blows only that branch's fuse without affecting relay logic, the kill switch path, or any other subsystem. Fuse ratings are selected above each circuit's expected steady-state draw but below the wire amperage rating, ensuring the fuse protects the wiring.

Chapter 4

Manufacturing

4.1 Hull

The hull was manufactured using a carbon fiber resin infusion process. It was split into three layups: the bottom half, the top half, and the canopy. A large piece of flat stock was also manufactured to be used for a deck door and bulkheads. The goal was to make a structural yet lightweight hull to handle the added weight of a payload at high speeds in choppy waters. Resin infusion was chosen for its cost-effectiveness, producing high-quality parts with good surface finish and water resistance. This section outlines the full manufacturing process from the design and fabrication of the molds to the final assembly of the hull.

4.1.1 Molds

A separate mold was designed for each of the three layups. The initial mold profile was created by subtracting the hull from a solid block to form a negative mold. This model was then split into top and bottom halves such that each could be manufactured as a single mold without overhangs. The split plane was defined through the sponson tips and the tangential edge of the transom.

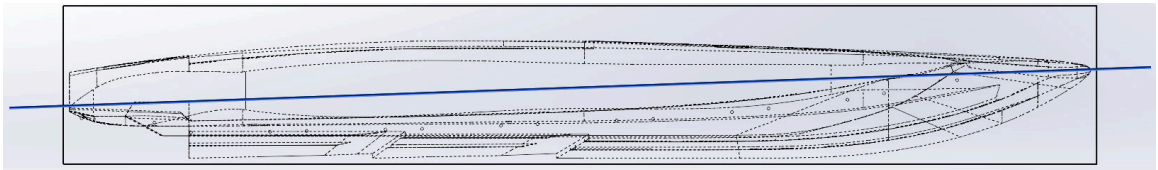


Figure 4.1: Plane where mold was cut in half

To eliminate overhangs in both halves, the sponson sides were normalized around their perimeter (Figure 4.2). While this slightly altered the hull geometry, it should have a negligible effect on performance.

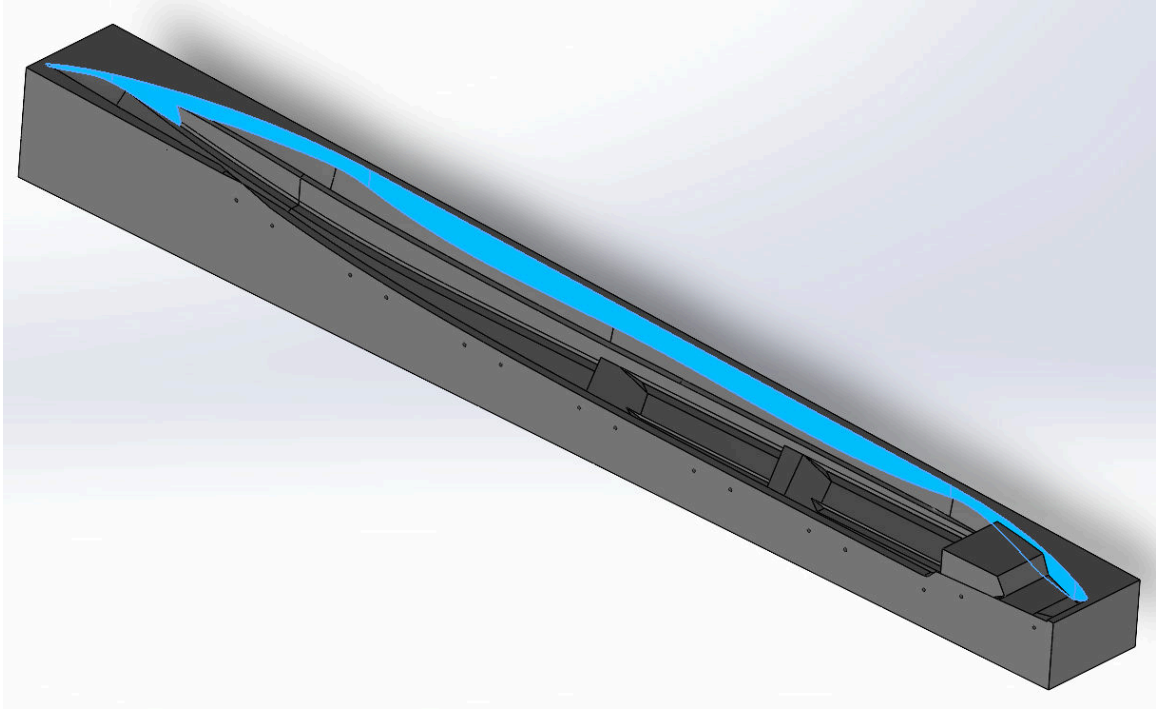


Figure 4.2: Flat Sponson Edge (Highlighted in blue)

The molds were fabricated from modular 3D-printed sections that were bolted together. In SolidWorks, the negative mold profiles were split up into roughly 9 inch sections, and tabs were added to allow the different sections to be bolted together. Each piece also had a flange that bolted to it. The purpose of the flange was to provide extra room for all the bagging material needed for the infusion. Each tab was designed with a hex indent in it for a 1/4 - 20 nut to simplify assembly.

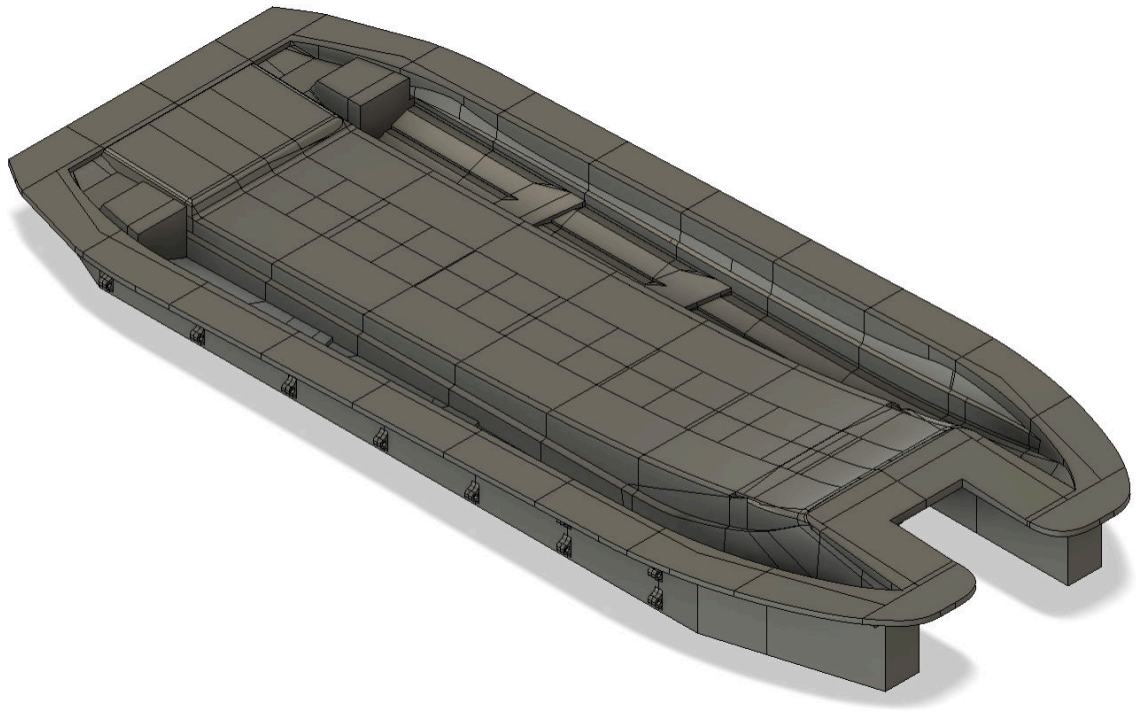


Figure 4.3: Bottom Half Mold CAD

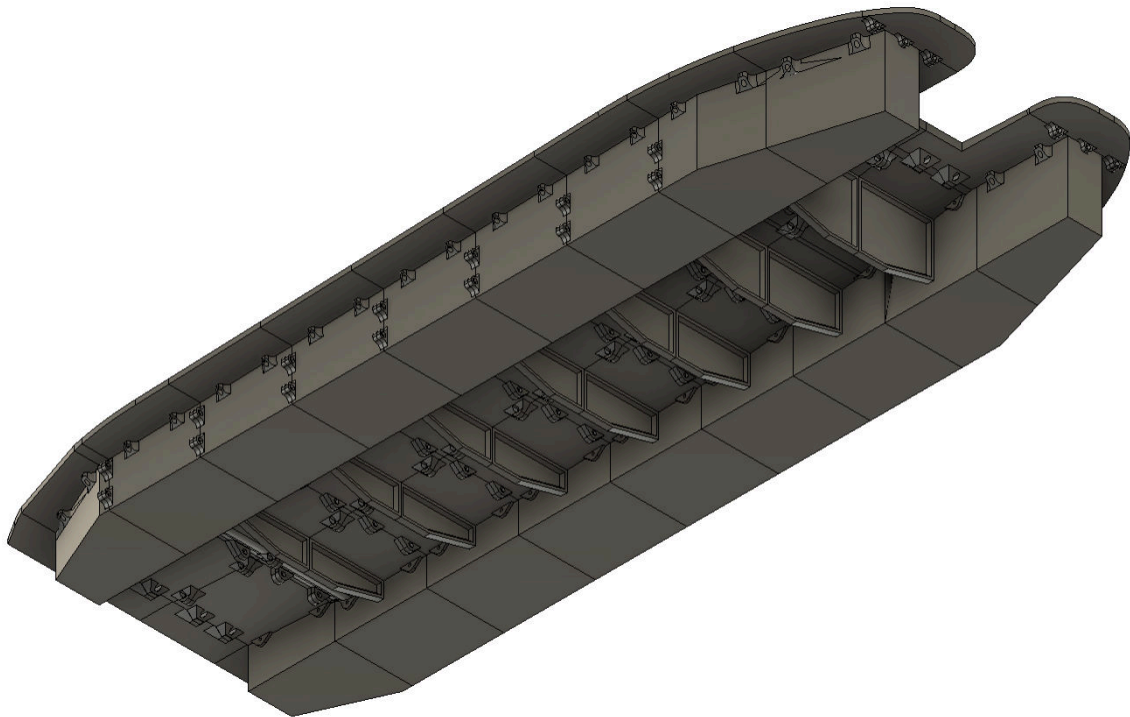


Figure 4.4: Bottom Half Mold CAD Bottom View



Figure 4.5: Top Half Mold CAD

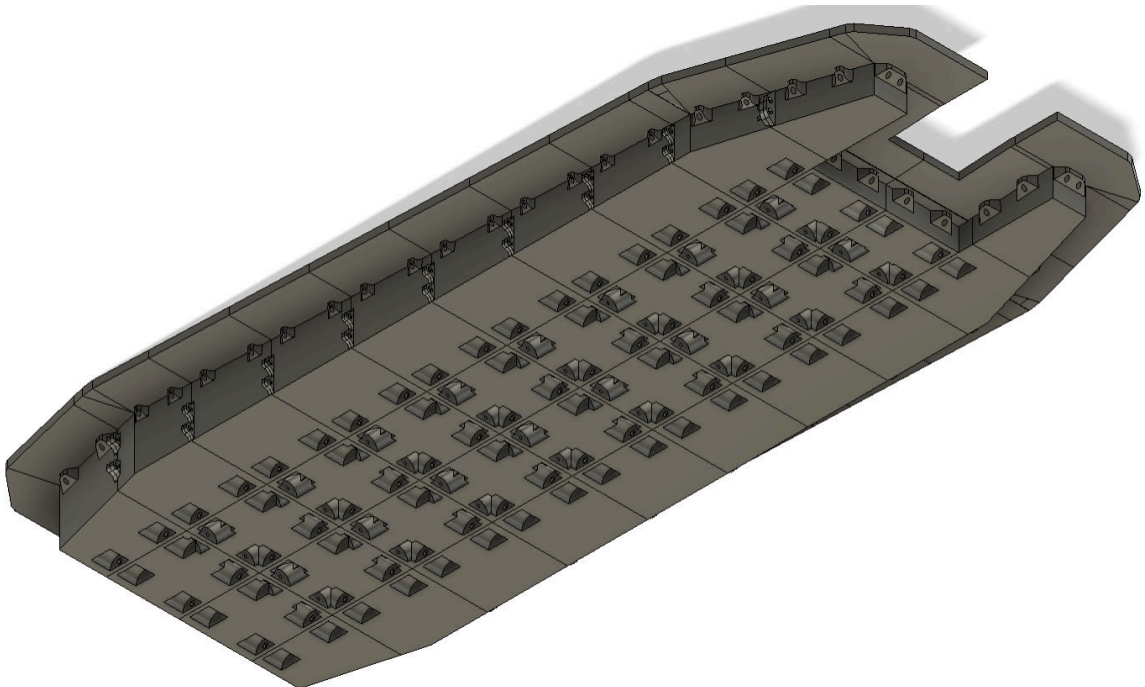


Figure 4.6: Top Mold CAD Bottom View

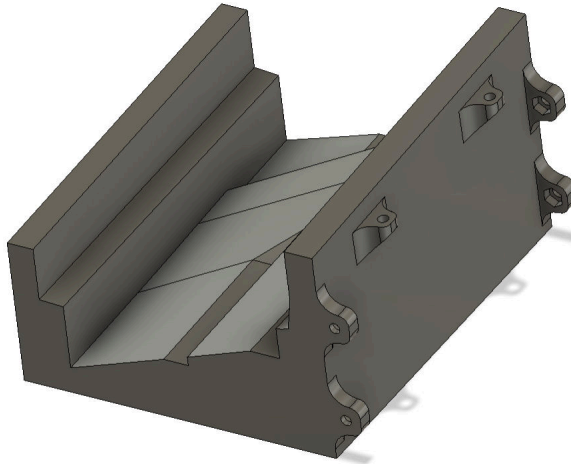


Figure 4.7: Example Mold Piece

Each mold piece (Figure 4.7) was 3D printed out of PLA with 2 walls and 5% infill. This configuration gave it enough structure to withstand surface prepping, while still conserving filament since each mold was going to require over 15 kg. These 3D printed pieces were then assembled into their respective molds. The seams between the prints were sealed with either Bondo or Liquid Nails. 2x4 lumber was also adhered to the underside of both molds to prevent them from flexing when being moved around. Both molds were sanded, sprayed with primer, and then wet sanded up to 2000 grit in order to ensure a good surface finish. Strips of Eucalyptus board were adhered around the edge of each mold to extend the flange. This board was sealed with caulk to prevent leaks when under vacuum.



Figure 4.8: Assembled Bottom Half Mold



Figure 4.9: Top and bottom molds assembled with 2x4 reinforcement



Figure 4.10: Top and bottom molds primed and with flange extension (photo taken after infusion part removal, so flange appears damaged)

The canopy mold, being much smaller, did not require the structural reinforcement. Instead of Bondo or Liquid Nails, thickened epoxy glue was used to seal the seams. Thin epoxy was coated over the whole mold in place of primer since epoxy does a better job sealing the mold while still being sandable. Since it was a simpler part, it didn't require the flange extensions either.

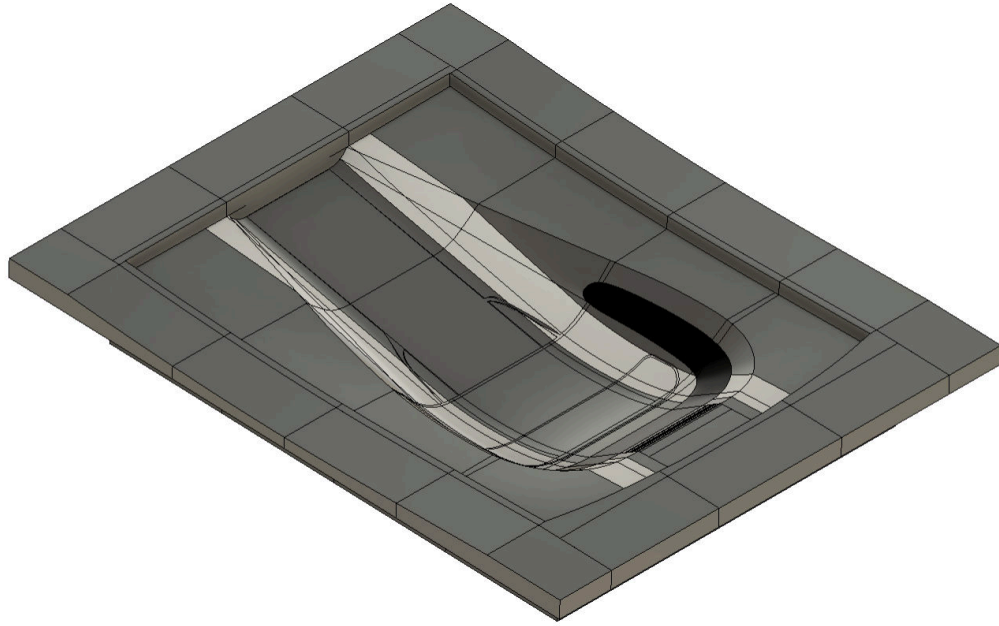


Figure 4.11: Canopy Mold CAD

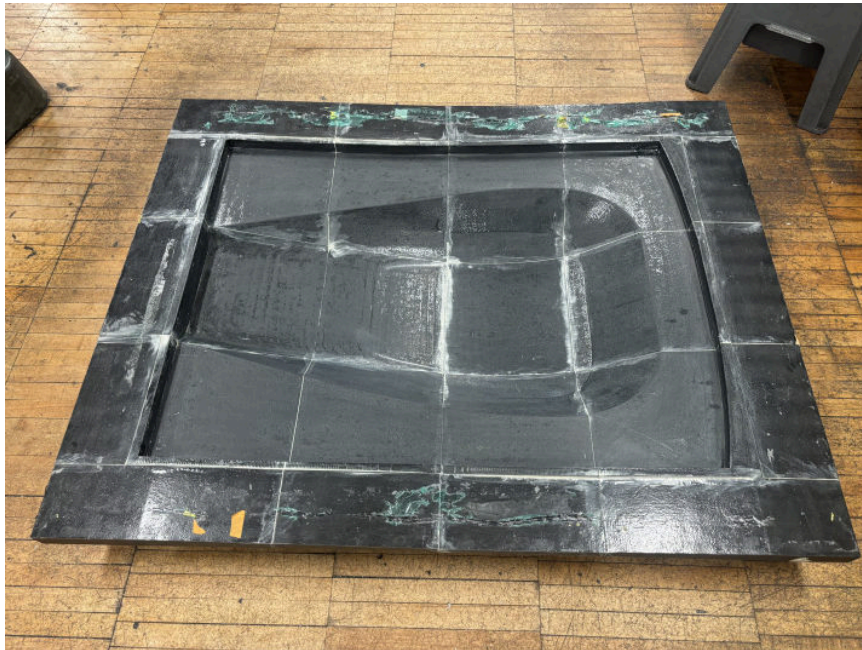


Figure 4.12: Completed Canopy Mold

4.1.2 Resin Infusion

The molds were ready for the carbon fiber resin infusion. Resin infusion is a composite manufacturing method in which carbon fiber is laid onto a mold and put under a vacuum before resin is allowed to flow into the part. It requires several different layers of material, but it is a cheap way to make a high-quality one-off part such as ours. The primary advantage of resin infusion is the ability to pull vacuum before the resin is mixed. Once the resin is mixed, there is limited time to get the part infused before the resin cures. Without that time constraint, the vacuum bag can be thoroughly inspected for leaks, ensuring a good vacuum pressure is achieved before letting the resin in.

The first step in the process is applying mold release. We elected to double-layer Partall Paste #2, a mold release wax, and Polyvinyl Alcohol (PVA). While the PVA on its own is likely enough, applying the wax first gives extra assurance that the part come out of the mold. Three coats of the Partall Paste were applied first in a wax-on wax-off manner. Then, four coats of PVA were painted on. For the canopy mold, we elected to do five coats of PVA and skip the wax.

After mold release, the next step is to dry lay the carbon. When laying down the carbon, we made sure the weave aligned with the length of the boat for aesthetic purposes. All three parts were made with 6 *oz/yd*² twill weave carbon fiber. For the bottom half, three layers were used since it bears all the load. The top half and canopy were both made with two layers since they didn't need to be as structural. In areas with hard corners or otherwise awkward shaping, the carbon was cut to be able to either lie around or double-layer over the area. These spots typically received an extra layer of carbon to patch anywhere that did not get full coverage. Load-bearing surfaces, such as the rear sponson blocks on the bottom half or the transom on the top half, received extra layers for reinforcement. To get the carbon to hold the shape of the mold when laying it down, a light spray adhesive was used. This spray does not affect the final surface finish.

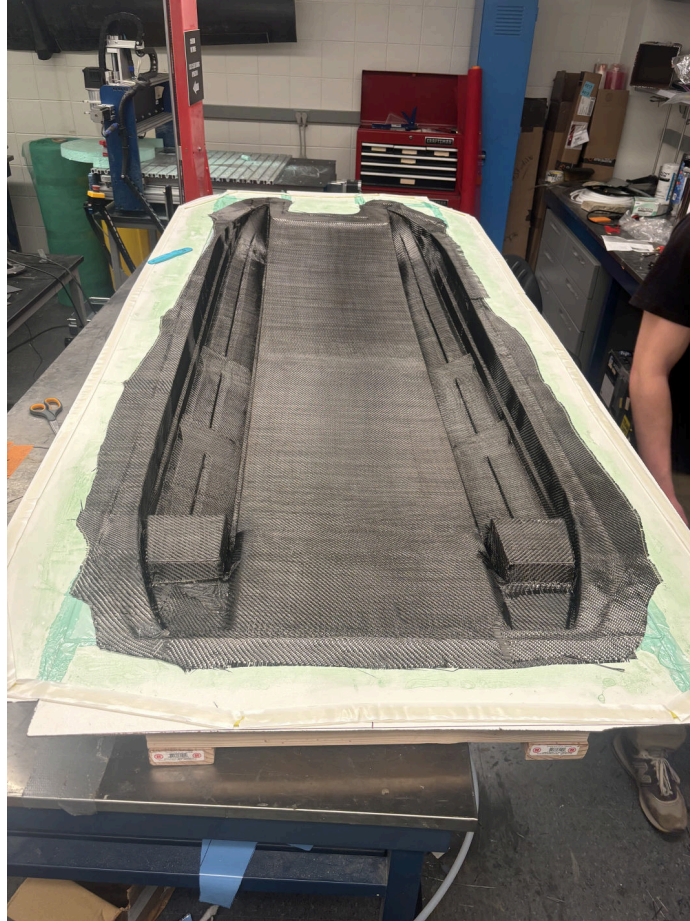


Figure 4.13: Carbon Laid Bottom Half Mold

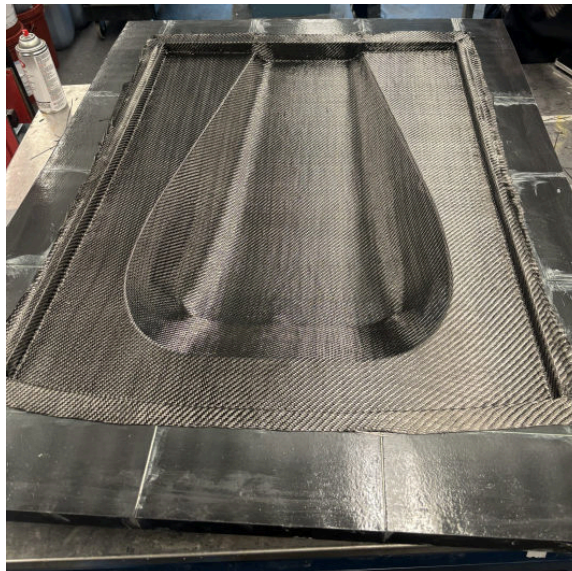
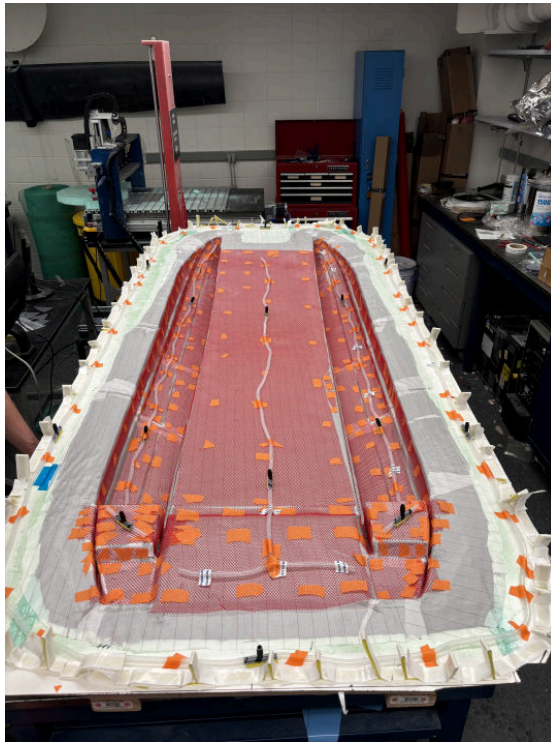


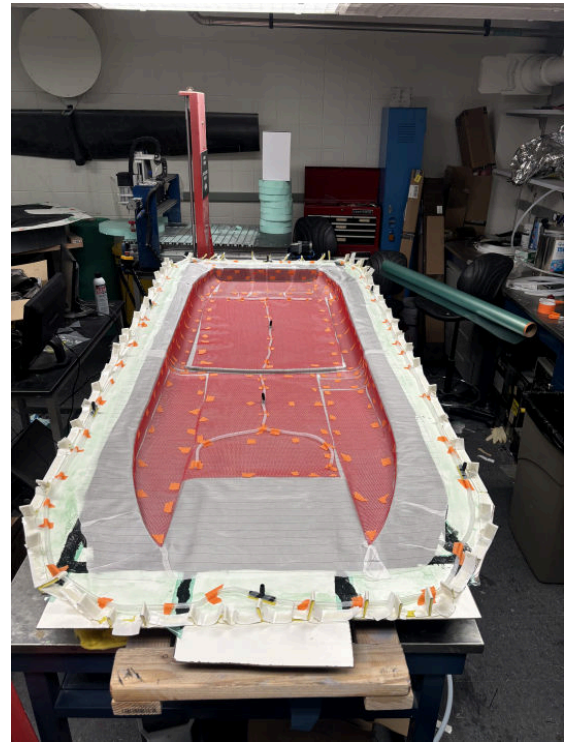
Figure 4.14: Carbon Laid Canopy Mold

For the top half of the mold, a deck door plate was put into place on the bare mold before the carbon was laid over it. The plate was cut from a piece of flat stock and was covered in mold release wax. It was held in place by a dab of caulk. Placing this deck door in before infusing the top results in a perfectly shaped indent being made in the top of the boat, seen in Figure 4.25.

The next layers are peel ply and flow media. Peel ply is a fabric layer that goes directly over the carbon. As its name implies, its purpose is to be more easily removable from the carbon when debuggging the part after infusion. Flow media is a stretchy plastic mesh whose purpose is to allow the resin to flow through the part. If flow media were not used, the vacuum would suffocate the part, and the resin would not be able to flow anywhere. Masking tape is used to hold the flow media down, but once again has no effect on the final part.



(a) Bottom Half



(b) Top Half

Figure 4.15: Fully layered molds prior to vacuum bagging

Bag tape and spiral chord are also shown in Figure 4.15. Bag tape creates an airtight seal between the mold and vacuum bag, while pleats provide excess material to conform to the mold geometry and prevent bridging. Spiral chord serves two roles in infusion: along the flange, it pulls the vacuum, and on the flow media, it supplies resin. Black T-fittings connect to vacuum or resin lines, enabling resin to

be introduced at the center and drawn outward. Resin lines remain pinched under vacuum and are opened sequentially during infusion.

After these steps, the molds are ready to bag. The bag is placed meticulously, making sure there are no creases along the bag tape that would cause leaks. Once the bag is down, the various hoses are connected and the vacuum is pulled. Then, the entire mold is inspected for leaks while also keeping an eye on the pressure gauge on the vacuum pump to see how close to a perfect vacuum it is. Leaks are plugged either with pieces of bag tape or with caulk.

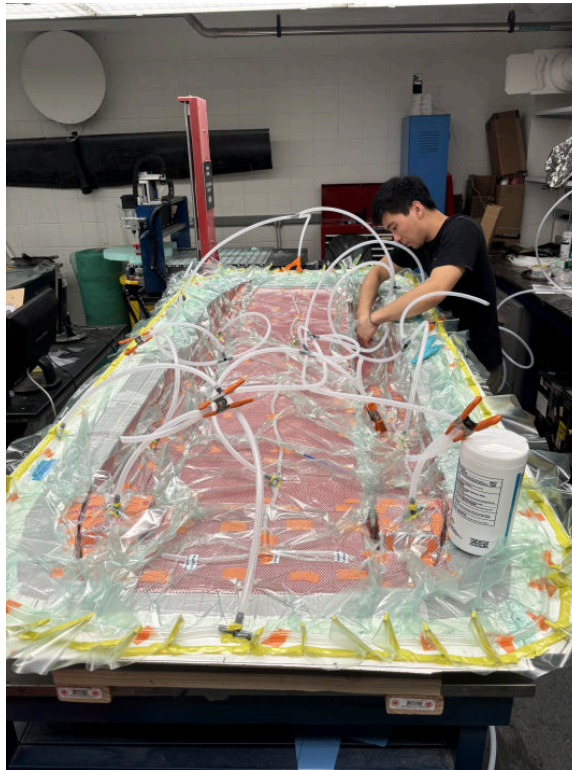
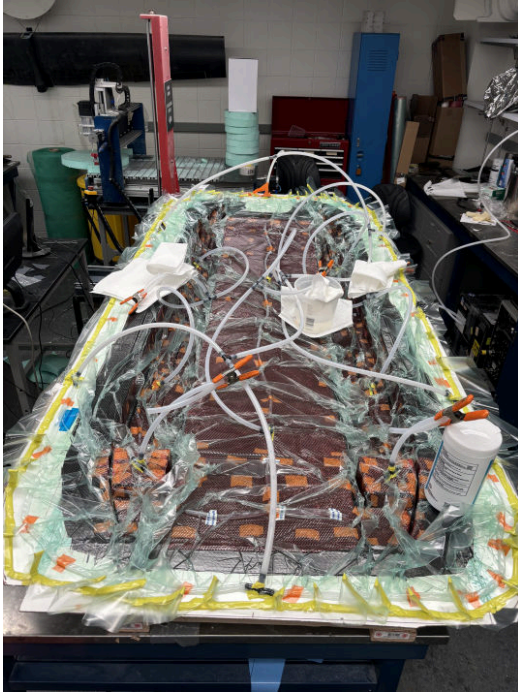
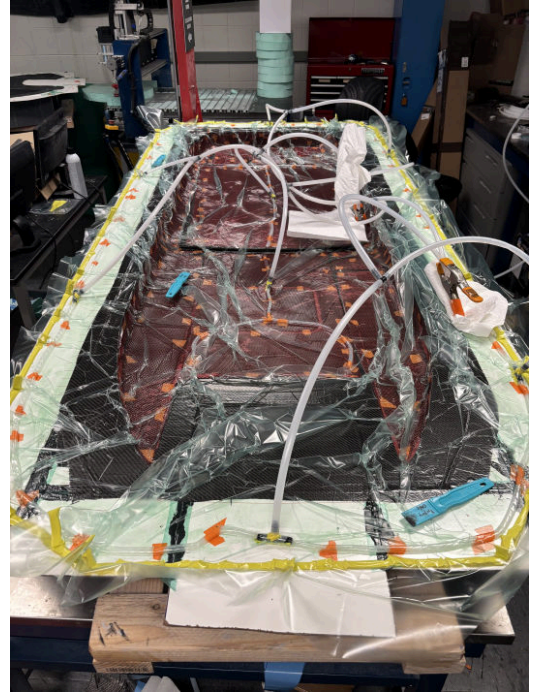


Figure 4.16: Bottom Mold Under Vacuum

Once the bag is securely under vacuum, resin is allowed to flow into the system. We elected to go with a slow-cure epoxy resin rather than a polyester or vinyl ester resin since epoxy is typically stronger. One at a time, the resin feed lines are opened with the end of the hose dipped in a cup of resin. For the bottom half, the feed line on the tunnel was opened first, followed by each of the sponsons. The feed lines on the rear sponson blocks didn't end up being necessary. The top half only had one connected feed line. Resin was fed into the system until all of the carbon was saturated, and then it was left to sit under vacuum for two days to let the epoxy cure.



(a) Bottom Half



(b) Top Half

Figure 4.17: Infused molds after resin infusion

With the epoxy fully cured, each layer of bagging material is pulled off until only the carbon remains. Then all that's left is to remove the finished part from the mold.



Figure 4.18: Bottom half being debagged



Figure 4.19: Finished composite parts in molds

4.1.3 Flat Stock

Before either half was infused, we also made a large piece of flat stock using a eucalyptus board as a mold. It was made with three layers of the same $6\text{ oz}/\text{yd}^2$ carbon. This served as a practice infusion, before the much more complicated hull infusions, as well as a stock to use for the deck door and bulkheads.

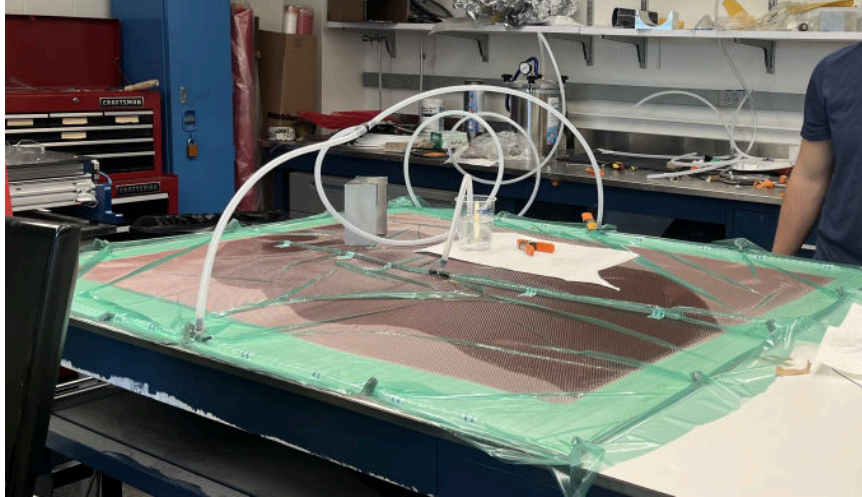


Figure 4.20: Flat Stock Infusion



Figure 4.21: Flat Stock

4.1.4 Post Processing

To remove the parts from the molds, wedges were inserted underneath the edge, and water was sprayed, or even siphoned, underneath (Figure 4.22). The mold release agent, PVA, is water-soluble, so getting water under the part dissolves the layer between the carbon and the mold, helping to release the part. This process would continue by pushing the wedges further under the part and adding more water until the part popped out of the mold.



Figure 4.22: Water being siphoned under the carbon to release part



(a) Bottom half raw part



(b) Top half raw part

Figure 4.23: Freshly removed composite parts

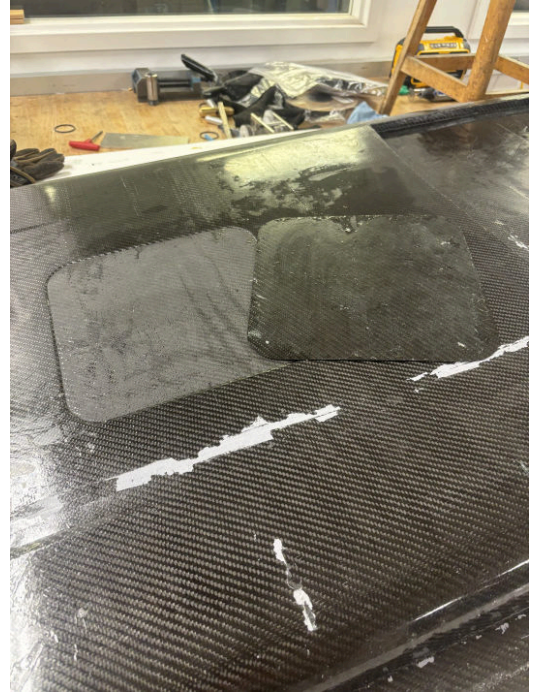
The flanges were then cut off and sanded smooth. The deck door was removed from the top half, and both it and the canopy hole were cut out. This provided ample accessibility to the inside of the hull when the two halves were assembled.



Figure 4.24: Bare Assembly



(a) Embedded deck door



(b) Embedded deck door removed

Figure 4.25: Deck Door Integration and Removal

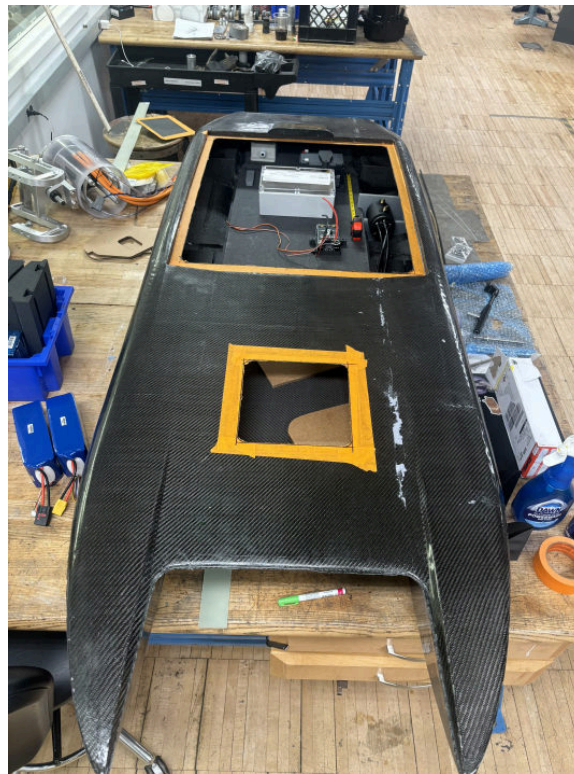
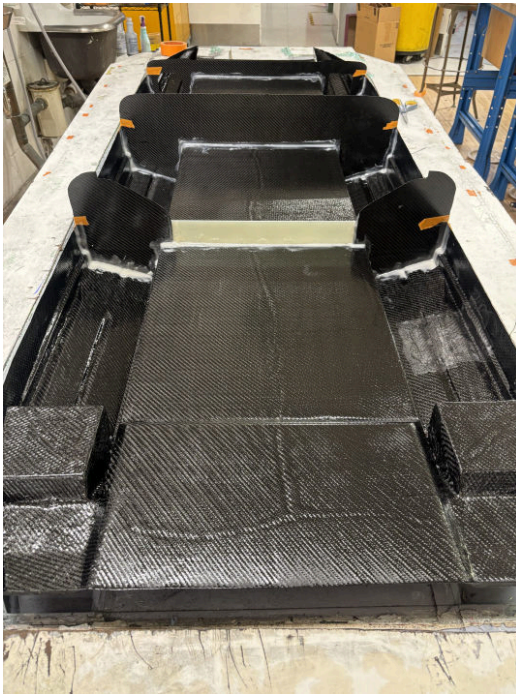


Figure 4.26: Assembled hull with deck doors cut

Both parts were coated internally with epoxy for waterproofing. Areas damaged during demolding or lacking sufficient carbon coverage were reinforced with one or two additional wet-laid carbon layers. A U-bolt was mounted under the front of the tunnel on the bottom half piece to tow the boat from, were it to cut out mid. This is also required per the PEP competition rules.

4.1.5 Assembly

The first part of the assembly process was cutting bulkheads out of the flat stock and gluing them to the bottom half. Stencils for the bulkheads were laser-cut out of cardboard and then traced onto the flat stock. Each bulkhead was then glued down using thick epoxy glue. The green spar across the tunnel in Figure 4.27a is a material called G10 fiberglass. It was added to prevent the tunnel from warping while also not interfering with cabling or hosing that needed to be routed over it during assembly. This same G10 material was used as a backing plate to reinforce several load-bearing parts of the hull, such as the transom. Along with the bulkheads, 3D-printed inserts were also epoxied into the front of the tunnel and sponsons in order to give the top half more surface area to bond to.



(a) Bulkheads glued down



(b) Bulkheads taped for bonding to the top half

Figure 4.27: Bulkhead Installation and Preparation

Before the top half was attached, the bottom half was tested for leaks (Figure 4.28). All the leaks found were marked and sealed with another coat of thin epoxy.



Figure 4.28: Float test of bottom half

To attach the two halves, the taped area at the top of each bulkhead in Figure 4.27b was filled with thick epoxy. Each of the 3D-printed inserts was also covered in thick epoxy. When the top half was placed on the bulkheads, the tape holding the epoxy made a fillet between the bulkhead and top half, creating a strong bond.



Figure 4.29: Top half gluing down to bulkheads

Finally, the gap between the top and bottom halves of the boat was sealed with fiberglass seam tape. This was wet laid all the way around the boat and then sanded smooth. The transom seam was also reinforced with more thick epoxy on the inside of the boat.



Figure 4.30: Fully assembled hull with seam tape

4.2 Driveline Assembly

The front motor support plate was printed with Bambu Lab PPA-CF, a nylon filament reinforced with carbon fiber. It provides strength comparable to metals, extreme temperature resistance, and dimensional precision and stability, making it a material suitable for replacing final automotive and industrial parts [34]. The plate is the main component transferring thrust load from the propellers to the vessel, so it is critical to use a material capable of withstanding compressive and bending loads. The remaining motor middle support and stuffing tube support plates are printed using ASA filament, which has sufficient strength properties but is not a critical property due to these plates taking negligible load.

Before final printing, test plates were printed to ensure proper fit into the correct locations in each sponson. The plates were designed by taking the contour of the sponsons from the 3D mold design at specified locations and lofting them to create a plate with a particular thickness. However, the CAD mold's contours differ from the actual carbon hull produced due to imperfections in the resin infusion process. As a result, multiple iterations were printed using PLA (Polylactic Acid) filament before a snug fit was achieved for each one. Notably, the "Press Pull" function on the Fusion 360 CAD software proved useful for quickly extending or reducing particular sections of the plates.

4.3 Component Mounts

Both styles of mounts previously discussed were 3D printed out of ASA. Prints were paused midway in order to insert stainless steel 1/4 - 20 Nylock nuts into the embedded nut cavities. Each print was then resumed, sealing the nuts away. All the hardware in the boat is stainless steel to prevent corrosion. Nylocks are used for all nuts to prevent bolts from loosening due to vibration. The base of each mount was attached to the hull via thick epoxy glue. Components were fastened to their respective mounts using 1/4 - 20 Hex Head bolts. Clamp-style mounts had longer bolts depending on the height of the component being mounted. The battery rails were printed out of ASA with 5 walls and 20% infill. The battery and payload trays, as well as the sliding nut inserts, were all printed out of ASA. Two 3"x3"x6" blocks of steel were machined to serve as the payload.

4.4 Steering Assembly

Both the servo mount and steering cable router were printed out of ASA and thick epoxy glued to the hull. An M4 rod was used to pin the sliding piece of the servo mount into position. The rudder mount was 3D printed out of PPA-CF since it is a load-bearing part. The rudder block was printed out of ASA and bolted to the rudder.

4.5 Low-Voltage Box

Figure 4.31 displays the assembled LV box. The fusebox was wired to two switches, which stick out of the side of the box: LV power and coolant. All of the wiring leaving the box is wired through a 12-pin Deutsch connector, which connects to the wiring harness on the outside.

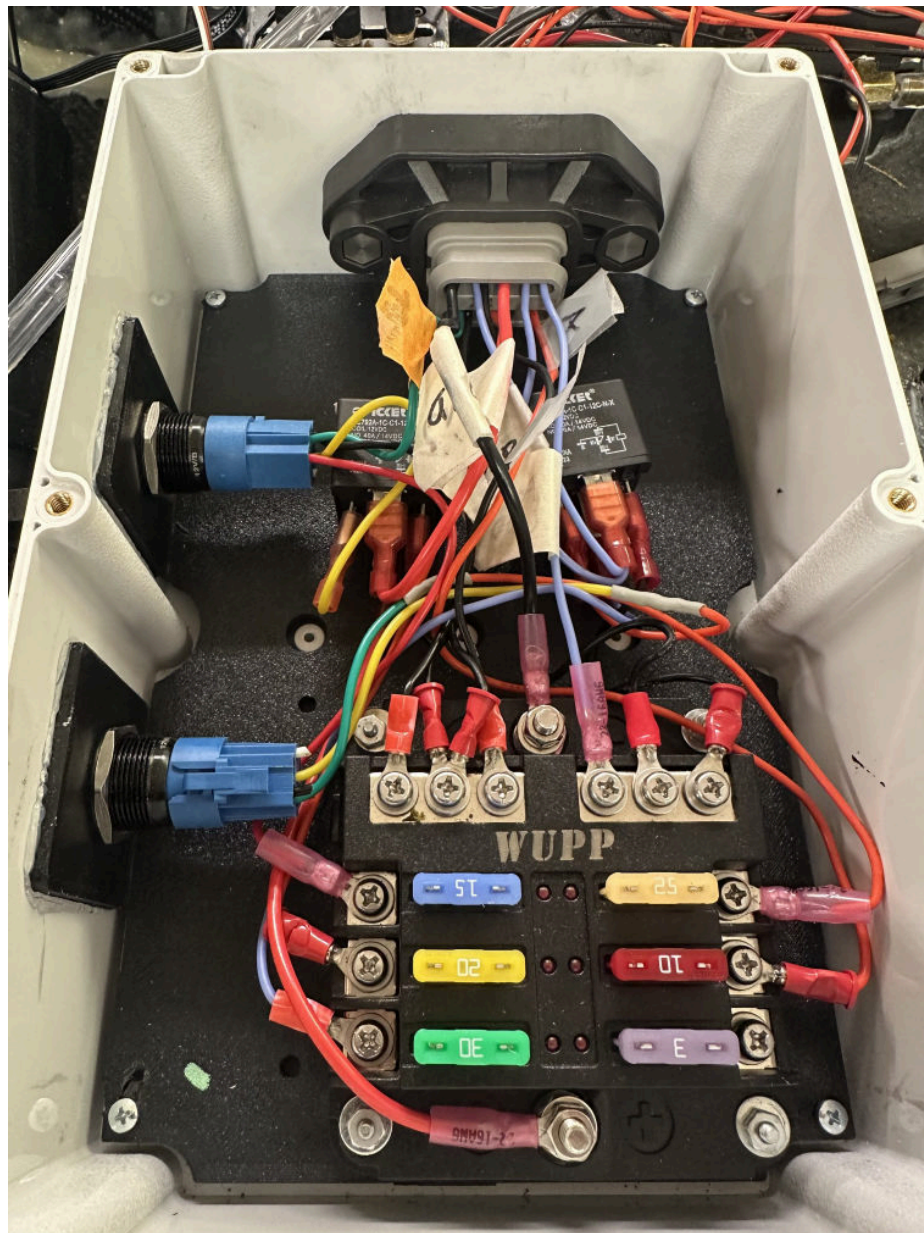


Figure 4.31: LV Box

Chapter 5

Integration

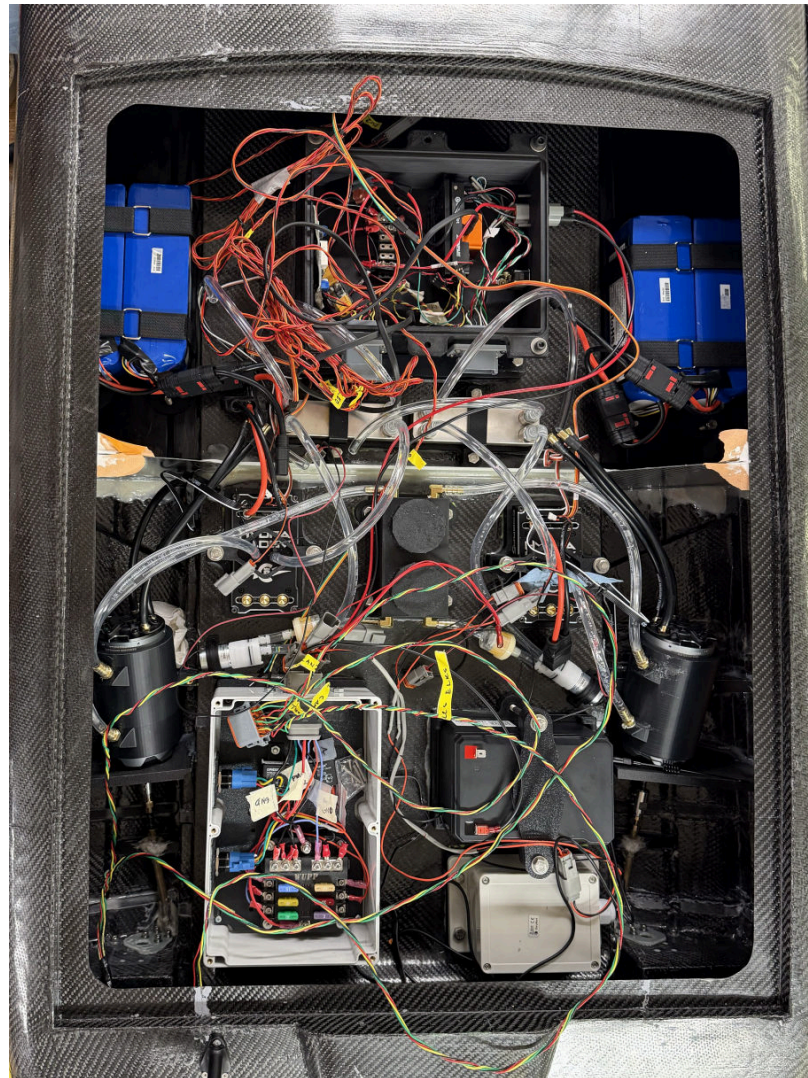


Figure 5.1: Fully Integrated System (No Payload)

Figure 5.1 shows the boat fully integrated with every component installed, excluding the payload. The configuration was chosen to optimize cable management, hose length, and workability.



Figure 5.2: Final Weigh In

As seen in Figure 5.2, the final weight of the boat, without payload, came in at 67.4 lbs. It is 22 lbs underweight compared to our initial estimate in Table 3.2. Most of this weight savings came from the hull. Our initial estimate for the hull weight was 12 lbs based solely on the carbon and resin weight that would be needed. However, we later increased our estimate to 40 lbs per advice received from a composites expert and hull design expert.

5.1 Mounting

Each mount was adhered to the hull via thick epoxy glue. Due to the imperfect shape of the hull, the battery rails required a lot of sanding to fit properly.

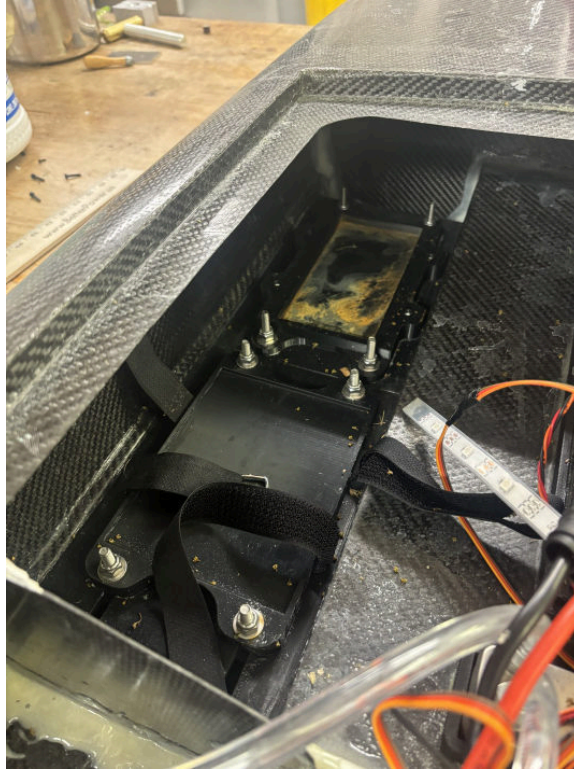
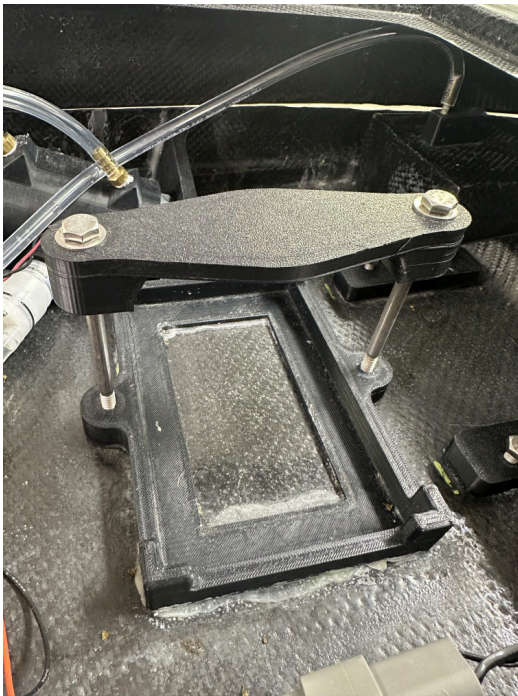


Figure 5.3: Battery Rails Installed



(a) LV battery mount (clamp style)



(b) Kill switch mount (direct bolt)

Figure 5.4: Both styles of mounting brackets installed

5.2 Coolant

Coolant was routed in the following order: reservoir, filter, pump, heat exchanger, ESC, motor, back to reservoir. This configuration ensures that the coldest water in the loop, right after the heat exchanger, goes to the most important component to cool, the ESC. Then it goes to the motor, the second most important component to cool. Raw water pickups were installed under the rear sponson blocks, routed through the heat exchangers, then out through a nipple on the side of the boat.



Figure 5.5: Water pickup (left) and coolant output nipple (right)

5.3 Steering

The servo was bolted to its mount with its horn attached. Teflon tubes were routed through the steering cable router and the rudder mount. The steering cable actuators were mounted to the rudder mount, which is bolted to the transom. All hardware that goes through the transom is coated in silicone to be waterproof. Silicone is also used to seal the back surface of the rudder mount to the transom. The steering cables were then fed through the actuators and teflon tubes. The rudder block was attached to the rudder and the steering actuators are fastened to the top of it. Finally, the steering cables were fed through the servo arms and tightened down.



Figure 5.6: Rudder Mount Integration



Figure 5.7: Integrated Steering Assembly

Chapter 6

Design Evaluation

6.1 Testing

Each system was tested individually before full assembly. The powertrain was tested under load using a load disk setup to evaluate how the ESCs and motors performed. The coolant system was run to test for leaks. The hull was float-tested multiple times to ensure water-tightness. After the boat was fully integrated, it was load-tested in Lake Carnegie with propellers installed. Finally, full system test runs were performed while gradually increasing range and speed. After each test, the data logs on the ESCs were checked for temperature and power draw through the Castle Link page shown in Figure 6.1.



Figure 6.1: Castle Link Data Log Page



Figure 6.2: Load Testing in Water



Figure 6.3: Full System Test Run

Throttle	Power-Our	Voltage	Ripple	Current	Temperati	Speed	Gov. Gain	BEC Volta	Motor Tem	Motor Tim	Acceleration.
1.9184	32.157	44.22	0.257	33	69.8	20030	0	7.35	73.4	2	0.31
1.9184	32.157	43.96	0.257	33	69.8	20000	0	7.35	73.4	2	0.25
1.9263	32.549	43.96	0	30	69.8	20100	0	7.35	73.4	2	0.25
1.9263	32.941	43.45	0.257	33	69.8	20290	0	7.35	73.4	2	0.13
1.9263	32.549	42.68	0.771	33	69.8	20250	0	7.35	73.4	2	0.19
1.9498	34.902	43.45	0.514	36	69.8	20950	0	7.35	73.4	3	0.19
1.9498	34.902	43.96	0.257	36	69.8	21100	0	7.35	73.4	3	0.19
1.9498	34.902	43.96	0.257	36	69.8	21140	0	7.35	73.4	3	0.13
1.9576	35.294	43.71	0	36	69.8	21320	0	7.35	73.4	3	0.19
1.9733	36.863	43.45	0.257	39	69.8	21380	0	7.35	73.4	3	0.25
1.9811	37.647	41.65	1.028	39	71.6	21890	0	7.35	73.4	3	0.25
1.9968	38.431	42.68	0.771	42	71.6	22420	0	7.35	73.4	3	0.19
2.0047	40	43.45	0.514	45	71.6	22910	0	7.35	73.4	3	0.19
2.0047	40	43.19	0.514	45	71.6	23150	0	7.35	73.4	3	0.25
2.0047	40	43.19	0.257	42	71.6	23230	0	7.35	73.4	3	0.19
2.0047	40	43.19	0.257	42	71.6	23320	0	7.35	73.4	3	0.19
2.0047	40	41.14	1.028	42	71.6	23340	0	7.35	73.4	4	0.13
1.9733	36.863	42.68	0.257	39	71.6	23130	0	7.35	73.4	3	0.13
1.8714	27.843	43.96	0.514	24	71.6	20960	0	7.35	73.4	2	-0.19
1.7696	18.824	44.74	1.028	15	71.6	19240	0	7.35	73.4	2	-0.19
1.6677	9.804	44.99	1.543	3	71.6	16670	0	7.35	73.4	1	-0.25
1.5658	1.176	45.25	1.543	0	71.6	11290	0	7.35	73.4	0	-0.31
1.5109	0	45.25	1.543	0	71.6	8950	0	7.35	73.4	0	-0.38
1.5109	0	45.51	1.543	0	71.6	8070	0	7.35	73.4	0	-0.31
1.5109	0	45.51	1.543	0	71.6	7710	0	7.35	73.4	0	-0.31
1.5109	0	45.51	1.543	0	71.6	7210	0	7.35	73.4	0	-0.38

Figure 6.4: Planing Run Data Log

Notable test data comes from the planing run where Osprey reached up to 30 mph, shown in Figure 6.3. The raw data log shown in Figure 6.4 shows peak current draw of 45 A, and max RPM of 23,150. The manufacturer lists max RPM as 38,000 and the Kv value as 680, meaning at 50.4 V, the max RPM is $680 \times 50.4 = 34,272$. We expected, with load and full throttle, the RPM range to be around 23-27,000 while pulling at least 100 A. However, the current draw only reached 45 A while reaching 23,150. This leads us to suspect the propeller diameter and/or pitch did not provide enough torque load for the motor to output power at its ideal RPM. However, it is difficult to know the ideal RPM range to run at without an existing motor datasheet. In addition, this run did not include the payload, which would increase the current draw due to increased weight.

Much more testing must be done in the future to fully validate the performance of Osprey.

6.2 Post-Assembly Challenges

Several challenges arose during the testing phase of this project. During the first several vessel float tests, hull leaks had to be located and sealed due to inconsistent resin infusion. An ESC caught fire during a dry ignition test, requiring additional time to repair damaged components in the vessel and to determine the root cause of the fire. At the PEP competition, a driveline misalignment had developed a day

before the official race. Finally, this misalignment caused the flex shaft to vibrate out of the shaft clutch collet and escape during on-water testing.

The first several water tests were intended for locating leaks throughout the hull. Leaks would be visually located and marked with tape while the hull floated in water. A layer of thin epoxy would be applied to the locations, and the hull would be rechecked for leaks. Fire damage from the ESC created an unnoticeable opening in the burned section of the hull, which also had to be resealed.

During a dry throttle test, when the kill switch circuit was closed to turn on the two ESCs via an ignition loop wired to the kill switch box, one ESC immediately began sparking and caught fire, before being put out with a fire extinguisher. The aftermath is shown in Figure 6.5. Having limited time to prepare the vessel for competition, immediate action was taken to clean everything of the extinguishing material and to determine the cause of the fire. Initial suspicion came from residual water shorting the positive and negative battery leads on the ESCs, but it could not explain the continual burning and smoking of the ESC. Looking at the data logs of the other ESC, the voltage ripple was logged as extremely high. Up to 27.767 V was measured in one session, or 60% of the battery-supplied voltage. We previously had no knowledge or understanding of voltage ripple, but online resources noted that maintaining ripple of under 10% is crucial to prevent ESC damage [35]. Other sessions frequently recorded ripple above 10%. Capacitors in the ESC help regulate these voltage fluctuations, but excessive amounts of damage to the Field Effect Transistors (FETs) can cause the ESC to fail. A solution to preventing ripple is to increase the discharge rating of the batteries, which is documented as a requirement for using the ESCs [36] [7]. This was recognized in the fall semester but regrettably overlooked. The selected batteries have a discharge rating of 12 C, significantly lower than the required 50 C rate the ESCs ask for. Other potential root causes include undersized connectors and wires, but those have been properly sourced to handle the expected continuous current. As a quick pivot, eight Lectron Pro 22.2V 5200mAh 100C Lipo batteries were purchased [37]. With a 2S-2P configuration per ESC, a nominal voltage of 44.4 V and capacity of 10400 mAh would be achieved. Since using the new batteries, the voltage ripple has been at safe levels, confirming the root cause was the low discharge rate.

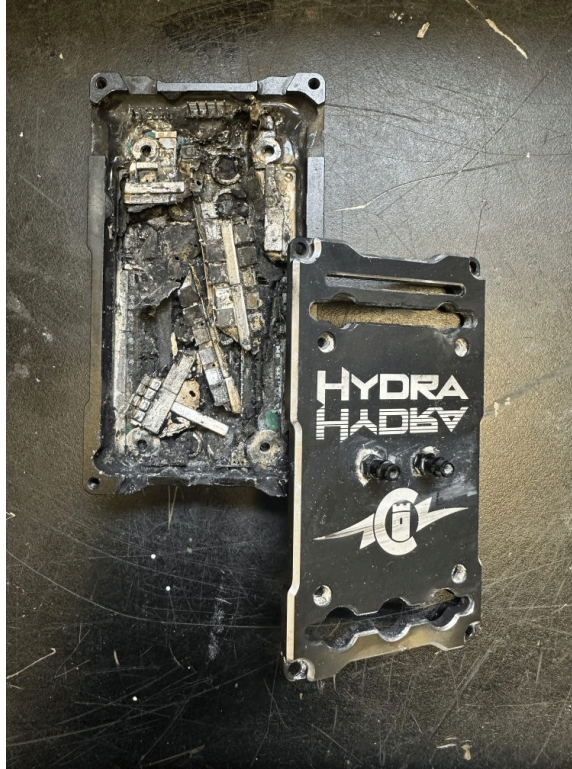


Figure 6.5: ESC after fire

Two days before the PEP competition race day, the vessel began having trouble steering, spinning in a circle despite turning the rudder the other direction. When eventually returning the vessel to shore, a physical and audible vibration in one of the drivelines was noticed. The shaft collet would also get hot, unbearable to touch with bare skin. When under load in the water, one propeller spun at a lower RPM, which was confirmed by ESC data logs. Because the drivelines had been running smoothly up to this point, a shift in the motor or collet position was suspected. The motor and collet were unmounted and reassembled, temporarily stopping the issue. The next day, the same behavior arose more severely. Close visual inspection revealed the cause to likely be non-concentricity of the shaft clutch collet and the flex shaft due to wear of the collet's cylindrical clamping surface. This misalignment introduced “jumping” of the flex shaft, which decreased RPM when a given throttle input was provided, and caused a droning noise that worsens with increased RPM. Friction between the collet and the flex shaft is also likely, given that the collet would get hot after high-RPM throttle.

To qualify for the real race, the vessel was run at 4 MPH to buy time for the issue to be troubleshooted. Once qualified, the vessel was tested again at the same speed. However, once out in the middle of the river, the vessel was incapable of

steering. It was inevitably towed back to shore when it was discovered that the flex shaft, along with the propeller, had escaped the collet and fallen out of the boat. Because of a large opening in the stinger where the flex shaft should have been, a significant amount of water filled the rear of the vessel enough to half-submerge the motors. Upon inspection, the nut clamping the shaft into the collet had not backed out, meaning the flex shaft vibrated enough to overcome the clamping force of the collet.

Chapter 7

Conclusion & Future Work

7.1 Progress Summary and Assessment

This thesis set out to design, build, and validate an unmanned payload-carrying surface vessel. Over the course of the project, we successfully met this objective by delivering a fully functional vessel with a complete hull design and powertrain setup.

A 7-foot catamaran resin-infused carbon fiber hull was fully manufactured in-house with custom-built 3D-printed molds. The finished parts came out with limited surface treatment required and plenty of structure. The vessel was completely sealed from leaks and has no major issues requiring repairs. A full electrical system was successfully designed and assembled to integrate the motor-ESC pair, 12.8 V battery, 22.2 V battery packs, steering servo, remote kill switch, cooling pumps, and RC/autonomous system. The RC system operated successfully, providing stable wireless control and telemetry during on-water testing. Early on-water runs without a payload ran smoothly, reaching up to 30 MPH while on plane.

While all major project requirements listed in Chapter 1 were satisfied, the most significant challenge faced was a lack of sufficient testing. Although we were able to perform the minimum required tests to validate the system's powertrain performance, extensive testing was not completed due to time constraints. Specifically, through Castle Link software, the Castle motor and ESC combination offers accessible data logs of temperature readings to validate the cooling system's design, as well as RPM and power output information for propeller selection. Range testing should also be done to determine the relationships between distance, time, speed, and current/power draw for a given set of batteries. This was unclear during competition time, and draining the battery packs to 0% charge can cause permanent damage, which we did not want to risk. Finally, CG tests are necessary to understand the CG's effect on

hull performance in different wind and water conditions.

7.2 Recommendations for Future Work

Many valuable experiences emerged over the course of design, integration, and testing that point toward future directions for improving the build process and Osprey's current state.

During testing, the raw water pickups never consistently flowed water through the heat exchangers, meaning that heat was never consistently removed from the ESC's and motors. During competition, when temperatures would reach 90 degrees Fahrenheit on a sunny day, the ESC's and motors would rest at 40 degrees C. The manufacturer requires them to be under 82 degrees Celsius, which could easily be reached if the powertrain is pushed for an extended period of time. As a reliable cooling solution during testing, pumps can be installed to pull in raw water instead of worrying about reaching minimum speeds for hydrodynamic pressure.

Bending the stuffing tubes to the desired shape proved trickier than anticipated. Kinking of the tube is an easy mistake and difficult to repair. Also, due to the springback of the brass, the custom bending jigs alone did not permanently hold shape. It was later discovered that annealing the tubes in a filament dryer at 85 degrees C while held in the jig significantly reduced springback. However, the PLA-printed jigs would begin warping at this temperature. An easy alteration is to print them with ASA, which has a glass transition temperature of 95 to 105 degrees C, while PLA's is 60 to 65 degrees C [38]. Another potential solution is to use a manual tube bender tool. They are typically designed for well-defined bends at particular points, but careful, minimal bending throughout a tube could work to reach the desired shape.

Consideration has been made of the future use of other motors in Osprey, such as the LMT 7065 [39]. With a different shape, bolt mounting pattern, and shaft, a different front support plate would be needed. To accommodate, a slot could be epoxied into the sponson that allows various plates to be installed.

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Appendix A

A.1 MATLAB Scripts

A.1.1 Hull Sizing Script

```
1 % Hull Design Calculator
2
3 %% --- Hull Sizing ---
4 l = 7; % ft, total boat length
5 c = (37.699/43.674)*1; % ft, chord/tunnel length
6 t = (3.305/37.699)*c*12; % in, wing thickness
7 h = (3.081/37.699)*c*12; % in, tunnel height at trailing edge,
   measured to chord line at back of boat, not back of
   sponsons
8 w = 18; % in, tunnel width
9 Deadrise_Angle = 21.5; % degrees
10 b = (2.892/43.674)*1; % ft, sponson pad width
11 Deck = ((12.511 - 6)*12/43.674)*1 + w; % in, deck width
12 haflchord = (10.24/37.699)*c; % ft, distance between transom
   and half chord length
13
14 %% --- Design Parameters ---
15 Vmph = 80; % mph, design speed
16 V = Vmph * 1.467; % ft/s, designed speed in ft/s
17 a_aero = 1.8; % Degrees, wing angle of attack with respect to
   bootom of hull
18 a_aero_rad = deg2rad(a_aero); % Radians, a aero
19 a_hydro = 1.45; % Degrees, angle of rear sponson step
20 a_hydro_rad = deg2rad(a_hydro); % Radians, a hydro
```

```

21 a_trim = 1.5; % Degrees, trim angle
22 a_trim_rad = deg2rad(a_trim); % Radians, a trim
23 a_wing_deg = a_aero + a_trim; % Degrees, Wing aoa
24 a_wing_rad = a_aero_rad + a_trim_rad; % Radians, Wing aoa
25 a_sponson_deg = a_hydro + a_trim; % Degrees, sponson aoa
26 a_sponson_rad = a_hydro_rad + a_trim_rad; % Radians, sponson
    aoa
27
28 %% --- Normalized Dimensions ---
29 t_ratio = (t/12) / c; % Thickness ratio
30 h_ratio = (h/12 + haflchord * sin(a_wing_rad)) / c; % Height
    ratio
31 AR = (w/12) / c; % Aspect ratio
32
33 %% --- Powertrain Parameters ---
34 Wtot = 130; % lbs, total weight
35 P_tot = 12*1.341; % Hp, total engine power
36 P_shaft = 0.9 * P_tot; % Hp, power at shaft (10% loss)
37 P_avail = 0.8 * P_shaft; % Hp, power out the prop (10% loss)
38
39 %% --- General Parameters ---
40 p_air = 0.00237; % slugs/ft^3, air density
41 p_water = 1.94; % slugs/ft^3, water density
42
43 %% --- Aerodynamic Lift ---
44 S = (w / 12) * c; % ft^2, lifting area
45 CLA1 = input(sprintf('Enter CLA1 value from Graph 1 for aoa =
    %.1f and hc = %.3f: ', a_wing_deg, h_ratio));
46 FAR = (AR - 0.25)/0.25 + 1; % CLA modification for aspect
    ratio
47 FTC = ((t_ratio - 0.07)/0.07 + 1) * 0.45; %
48 Cl = CLA1 * FAR * FTC; % Lift coefficient
49 LA = aeroForce(p_air, V, S, Cl); % lbs, total aerodynamic lift
50
51 fprintf("\n=== AERODYNAMIC LIFT ===\n");
52 fprintf("Lift Coefficient (Cl): %.3f\n", Cl);
53 fprintf("Lifting Area (S): %.3f ft^2\n", S);
54 fprintf("Aerodynamic Lift (LA): %.3f lbs\n", LA);

```



```

55
56 %% --- Aerodynamic Drag ---
57 CDI = input(sprintf('Enter CDI value from Graph 2 based on
    CLA1 = %.3f and h_ratio = %.3f: ', CLA1, h_ratio));
58 CDO = (0.055/0.07) * t_ratio; % Friction drag coefficient
59 CDA = CDI + CDO; % Total drag coefficient
60 Savg = ((w / 12) + (Deck / 12)) / 2 * c; % ft^2, average drag
    area
61 DA = aeroForce(p_air, V, Savg, CDA); % lbs, total aerodynamic
    drag
62
63 fprintf("\n=== AERODYNAMIC DRAG ===\n");
64 fprintf("Induced Drag Coefficient (CDI): %.3f\n", CDI);
65 fprintf("Friction Drag Coefficient (CDO): %.3f\n", CDO);
66 fprintf("Total Drag Coefficient (CDA): %.3f\n", CDA);
67 fprintf("Average Drag Area (Savg): %.3f ft^2\n", Savg);
68 fprintf("Aerodynamic Drag (DA): %.3f lbs\n", DA);
69
70 %% --- Required Hydro Lift ---
71 LW = Wtot - LA; % lbs, required hydrodynamic lift
72 fprintf("\n=== HYDRODYNAMIC LIFT ===\n");
73 fprintf("Total Weight (Wtot): %.3f lbs\n", Wtot);
74 fprintf("Required Hydro Lift (LW): %.3f lbs\n", LW);
75
76 %% --- Wetted Area Iteration ---
77 S1 = .2; % ft^2, initial guess for wetted area
78 [Sw, CLW, iterations, history] = calcWettedAreaIterative(LW,
    p_water, V, b, S1, a_sponson_deg, Deadrise_Angle);
79
80 % Final geometry based on converged Sw
81 lw_final = Sw / (2 * b); % ft, wetted length
82 ARs_final = b / lw_final; % aspect ratio
83
84 fprintf("\n=== WETTED AREA RESULTS ===\n");
85 fprintf("Final Wetted Area (Sw): %.3f ft^2\n", Sw);
86 fprintf("Final Lift Coefficient (CLW): %.3f\n", CLW);
87 fprintf("Final Wetted Length (lw): %.3f ft\n", lw_final);
88 fprintf("Final Aspect Ratio (ARs): %.3f\n", ARs_final);

```

```

89 fprintf("Iterations to Converge: %d\n", iterations);
90
91 fprintf("\n--- Iteration History ---\n");
92 disp(history);
93
94 %% --- Hydrodynamic Drag ---
95 SAF = 0.5; % Spray area factor
96 vel_len = V*lw_final; % ft^2/sec;
97 CDF1 = input(sprintf('Enter CDF1 value from Graph 4 based on
    Velocity-Length = %.3f: ', vel_len)); % CDF 1 from Graph 4
98 CDF = CDF1*(1+SAF); % Friction drag coefficient accounting for
    SAF
99 CDIW = CLW*(0.0008/0.022)*(a_sponson_deg/2); % Induced drag
    coefficient
100 CDP = 0.0005; % Profile drag
101 CDW = CDF + CDIW + CDP; % Total drag coefficient due to water
102 DW = aeroForce(p_water, V, Sw, CDW); % lbs, total aerodynamic
    drag
103
104 fprintf("\n=== Hydrodynamic DRAG ===\n");
105 fprintf("Induced Drag Coefficient (CDIW): %.3f\n", CDIW);
106 fprintf("Friction Drag Coefficient (CDF): %.3f\n", CDF);
107 fprintf("Total Drag Coefficient (CDW): %.3f\n", CDW);
108 fprintf("Hydrodynamic Drag (DW): %.3f lbs\n", DW);
109
110 %% --- Appendage Drag ---
111 DM_full = input(sprintf('Enter DM value from Graph 5 for Group
    3 based on Velocity = %.3f: ', V)); % Race engine drag
112 DM = DM_full/5; % Scaled motor drag for RC size boat
113 Sa = (3.6810%*2)*(1/43.674)^2; % ft^2, Appendage area
114 D_cockpit = aeroForce(p_air, V, Sa, 0.3); % lbs, total
    aerodynamic drag
115 fprintf("\n=== Appendage DRAG ===\n");
116 fprintf("Appendage Drag (Dcockpit): %.3f lbs\n", D_cockpit);
117
118 %% --- Total Drag ---
119 D = DA + DW + DM + D_cockpit; % lbs, Total Drag
120 fprintf("\n=== Total DRAG ===\n");

```

```

121 fprintf("Total Drag (D): %.3f lbs\n", D);
122
123 %% --- Power Required ---
124 P_req = (D*V)/550; % Hp, power required to balance thrust and
    drag at design velocity
125 P_diff = P_avail - P_req; % Power Overhead
126
127 fprintf("\n== Power Requirement ==\n");
128 fprintf("Power Required (P_req): %.3f Hp\n", P_req);
129 fprintf("Power Overhead (P_diff): %.3f Hp\n", P_diff);

```

Listing A.1: MATLAB script for hull sizing calculations

A.1.2 aeroForce Function

```

1 function F = aeroForce(rho, V, S, C)
2     % aeroForce computes an aerodynamic force (lift, drag, etc
    .)
3     % using the general aerodynamic force equation:
4     %      F = 0.5 * rho * V^2 * S * C
5     %
6     % Inputs:
7     %   rho - fluid density (slugs/ft^3)
8     %   V   - velocity (ft/s)
9     %   S   - reference area (ft^2)
10    %   C    - aerodynamic coefficient (dimensionless)
11    %          (e.g., Cl for lift, Cd for drag)
12    %
13    % Output:
14    %   F    - aerodynamic force (lb)
15    %
16    % Example:
17    %   L = aeroForce(0.00237, 125, 60, 0.293); % lift
18    %   D = aeroForce(0.00237, 125, 60, 0.040); % drag
19
20    F = 0.5 * rho * V^2 * S * C;
21 end

```

Listing A.2: MATLAB function for lift and drag calculations

A.1.3 calcWettedAreaIterative Function

```
1 function [Sw, CLW, iterations, history] =  
    calcWettedAreaIterative(LW, p_water, V, b, S1, aoa,  
    deadrise)  
2     % calcWettedAreaIterative  
3     % Iteratively solves for the wetted area Sw until  
        convergence.  
4     % Prompts the user to input CLW for each iteration given  
        ARs and aoa.  
5  
6     tol = 1e-3;          % convergence tolerance  
7     maxIter = 50;        % safety limit  
8     iterations = 0;  
9  
10    % Initialize arrays for history tracking  
11    iterList = [];  
12    S1List = [];  
13    ARsList = [];  
14    CLWList = [];  
15    SwList = [];  
16  
17    FDR = (10-deadrise)/50 + 1; % CLW Modifier for dearise  
        angle  
18  
19    while true  
20        iterations = iterations + 1;  
21  
22        % Compute ARs based on current guess S1  
23        lw = S1 / (2 * b); % wetted length (ft)  
24        ARs = b / lw;      % aspect ratio of wetted pad  
25  
26        % Display iteration info  
27        fprintf('\nIteration %d\n', iterations);  
28        fprintf('Current ARs = %.4f, aoa = %.2f degrees\n',  
            ARs, aoa);  
29  
30        % User input for CLW
```

```

31     CLW_b10 = input('Enter CLW value for this ARs and aoa:
    ');
32     CLW = CLW_b10 * FDR; % CLW accounting for deadrise
    angle
33
34     % Compute new wetted area
35     Sw = LW / (0.5 * p_water * V^2 * CLW);
36
37     % Store iteration data
38     iterList(end+1) = iterations;
39     S1List(end+1) = S1;
40     ARsList(end+1) = ARs;
41     CLWList(end+1) = CLW;
42     SwList(end+1) = Sw;
43
44     % Check convergence
45     if abs(Sw - S1) < tol
46         fprintf('Converged after %d iterations.\n',
            iterations);
47         break;
48     elseif iterations >= maxIter
49         warning('Maximum iterations reached without
            convergence. ');
50         break;
51     else
52         S1 = Sw; % update for next iteration
53     end
54 end
55
56 % Create final table with requested columns
57 history = table(iterList', S1List', ARsList', CLWList',
    SwList', ...
58     'VariableNames', {'Iteration', 'S1', 'ARs', 'CLW', 'Sw
    '});
59 end

```

Listing A.3: MATLAB script for wetted area iteration